Chapter 6 Water Treatment

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Expected leaning outcomes for Chapter 6

- Water treatment
  - Treatment schemes: Surface water treatment; Ground water treatment
  - Sedimentation
  - Coagulation and flocculation
  - Filtration
  - Disinfection
  - Hardness, alkalinity, and softening methods
  - Membrane processes (MF, UF, NF, and RO)

Water is life



The New York Times

Russia-Ukraine War › | U.S.-Russia Talks | Ukraine's Ballistic Missile | A Harsh Winter | Remaking War at Sea | 'Fortress Cities'

Russia Is Planning Its Most Powerful Blow Yet to Try to Freeze Ukraine

Ukraine showed European partners what it said were intercepted Russian plans to cut off major cities from power, heat and water this winter.

The Guardian

Int

Sport | Culture | Lifestyle

Middle East | Ukraine | Environment | Science | Global development | Football | Tech | Business | Obituaries

This article is more than **3 months old**

Military strikes on water facilities in Iran may constitute a war crime, experts say

Strikes on Bemani damaged key water reservoir for 20,000 people living in area amid a historic drought in the country



War on Iran: The Dangers of Attacking Water Desalination Plants in the Gulf

Apr 28, 2026 | Manal Shehabi



International attention has focused on the energy and economic impact of Iran's effective closure of the Strait of Hormuz. Yet for people living in the Gulf Cooperation Council (GCC) states and Iran, the war also has the potential to jeopardize an even more vital resource: water. In this region, the most water-stressed globally, water desalination—turning saltwater into fresh water suitable for human consumption and other uses—is essential to sustaining lives, economies, and security. Since the United States and Israel launched their war on Iran on February 28, 2026, several desalination plants in the Gulf region have been hit, affecting the operation of water systems that are critical for civilian populations.

Water treatment systems

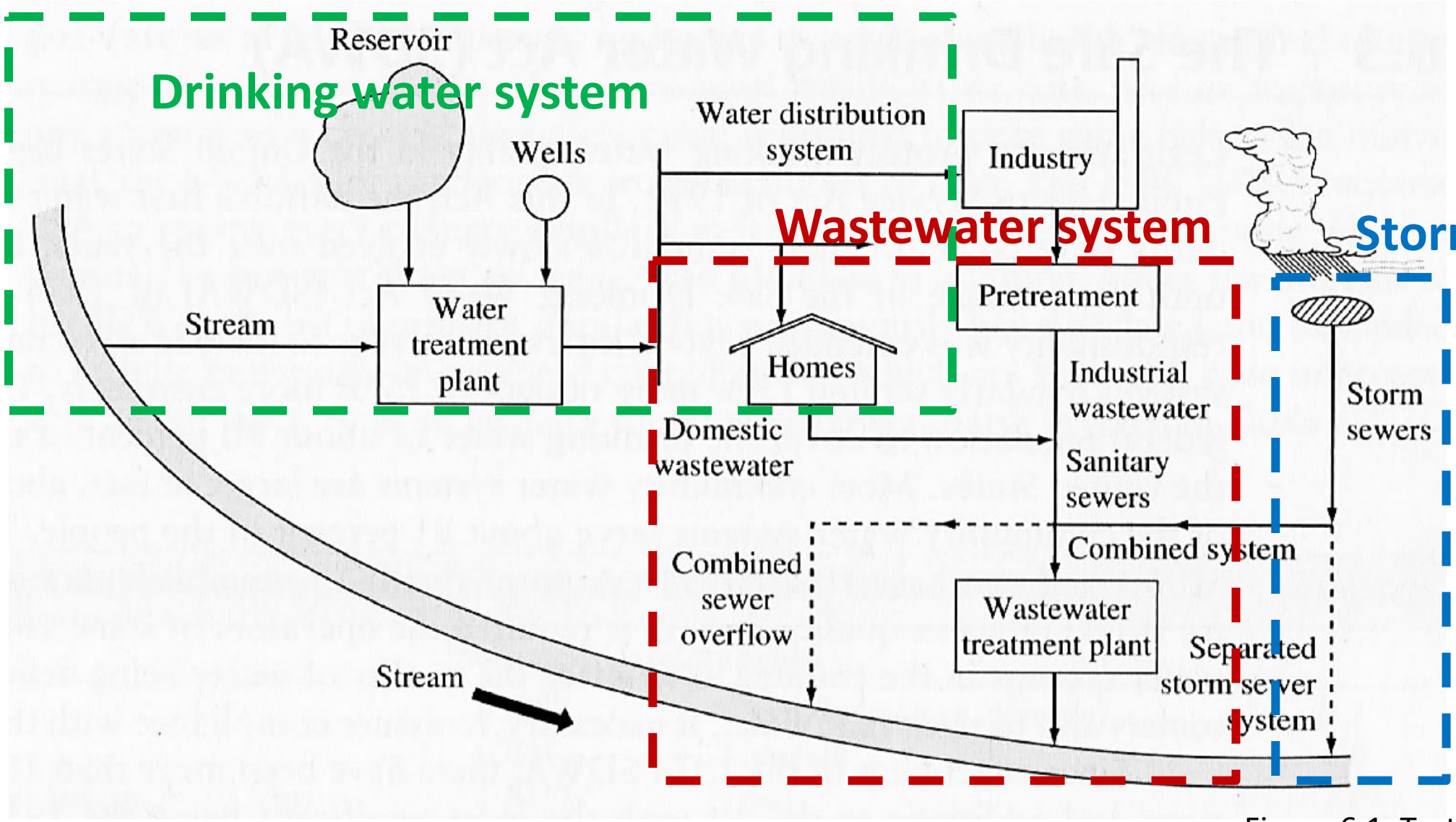
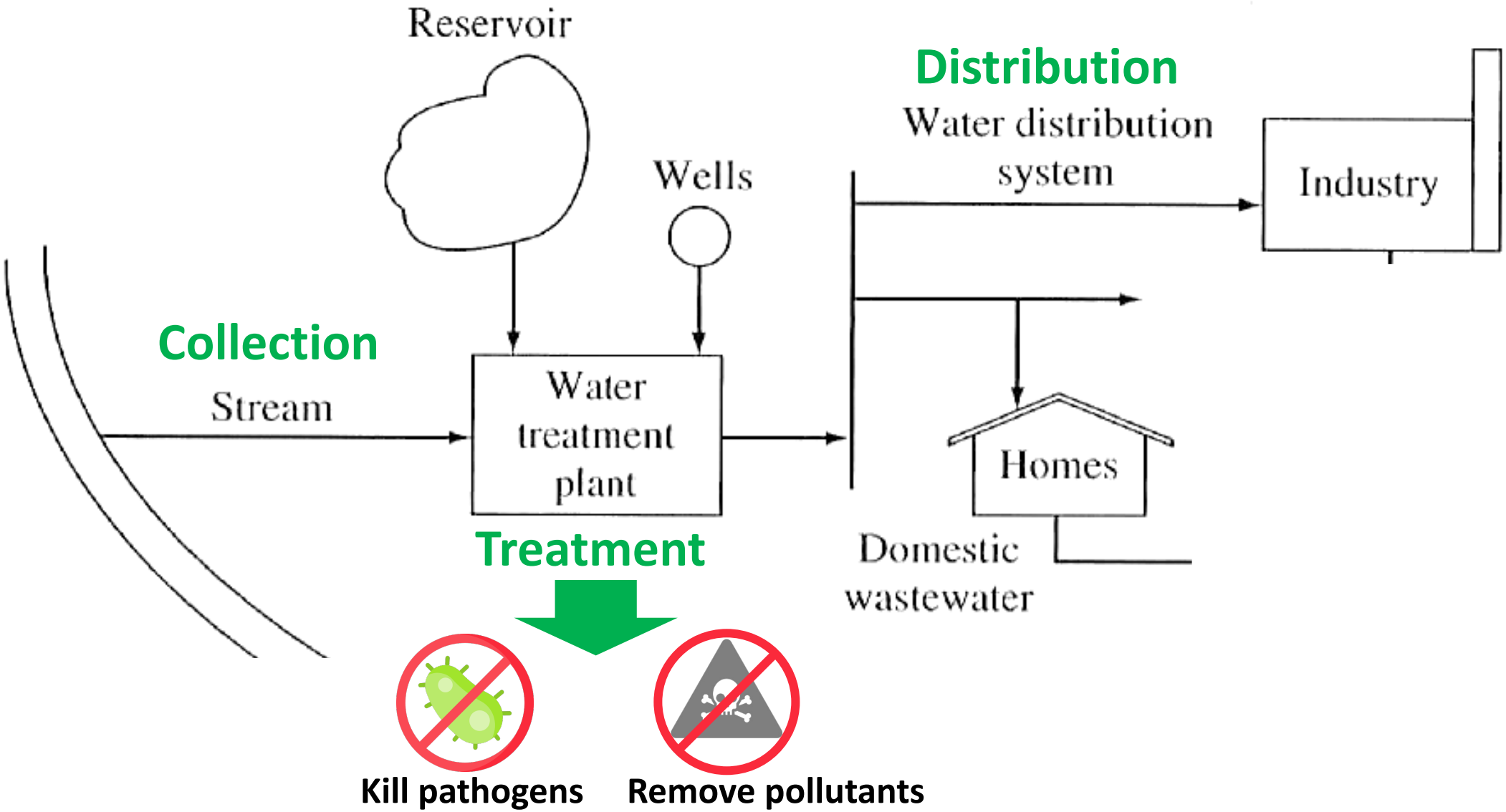
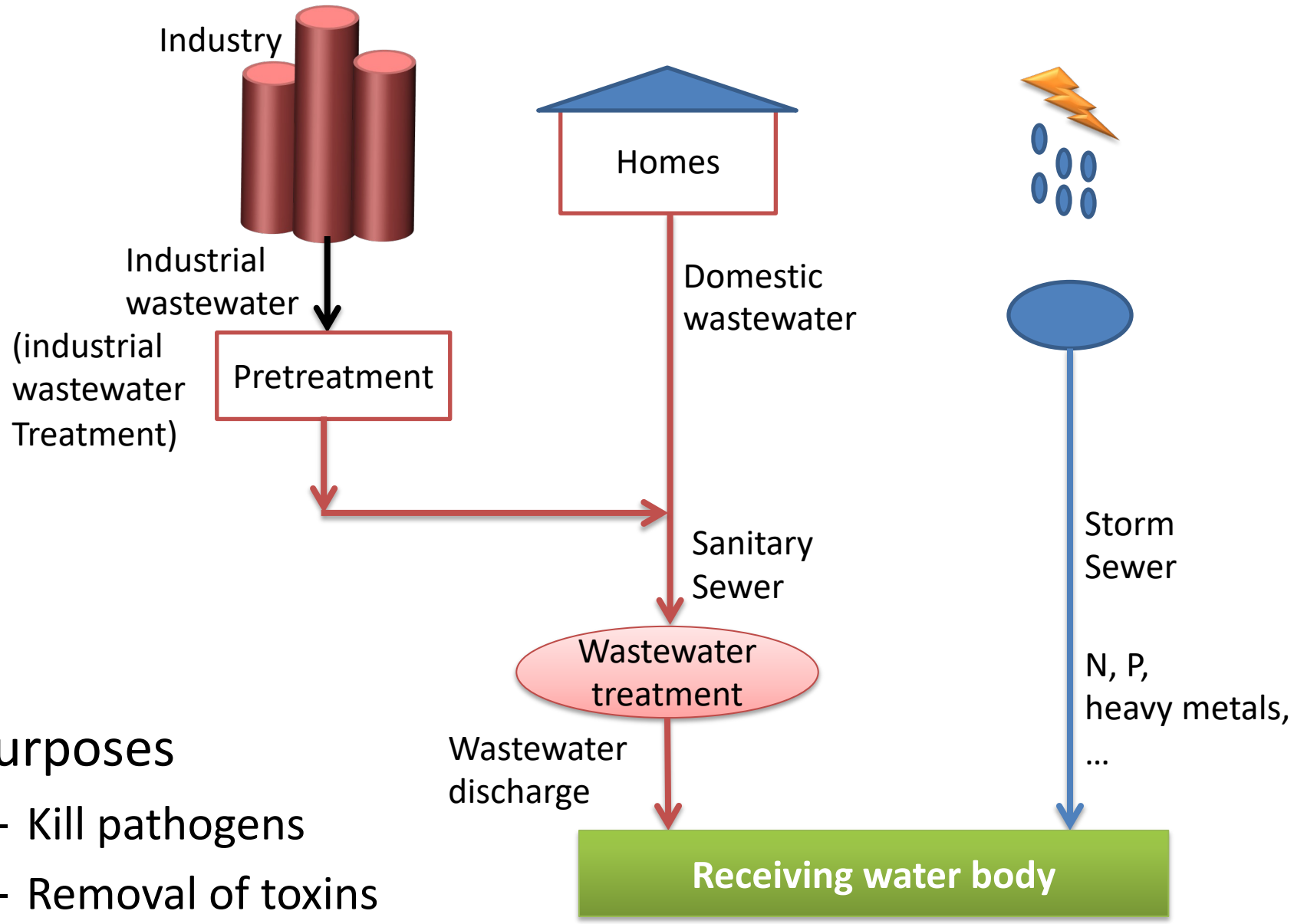


Figure 6.1, Textbook, p283

Drinking water system

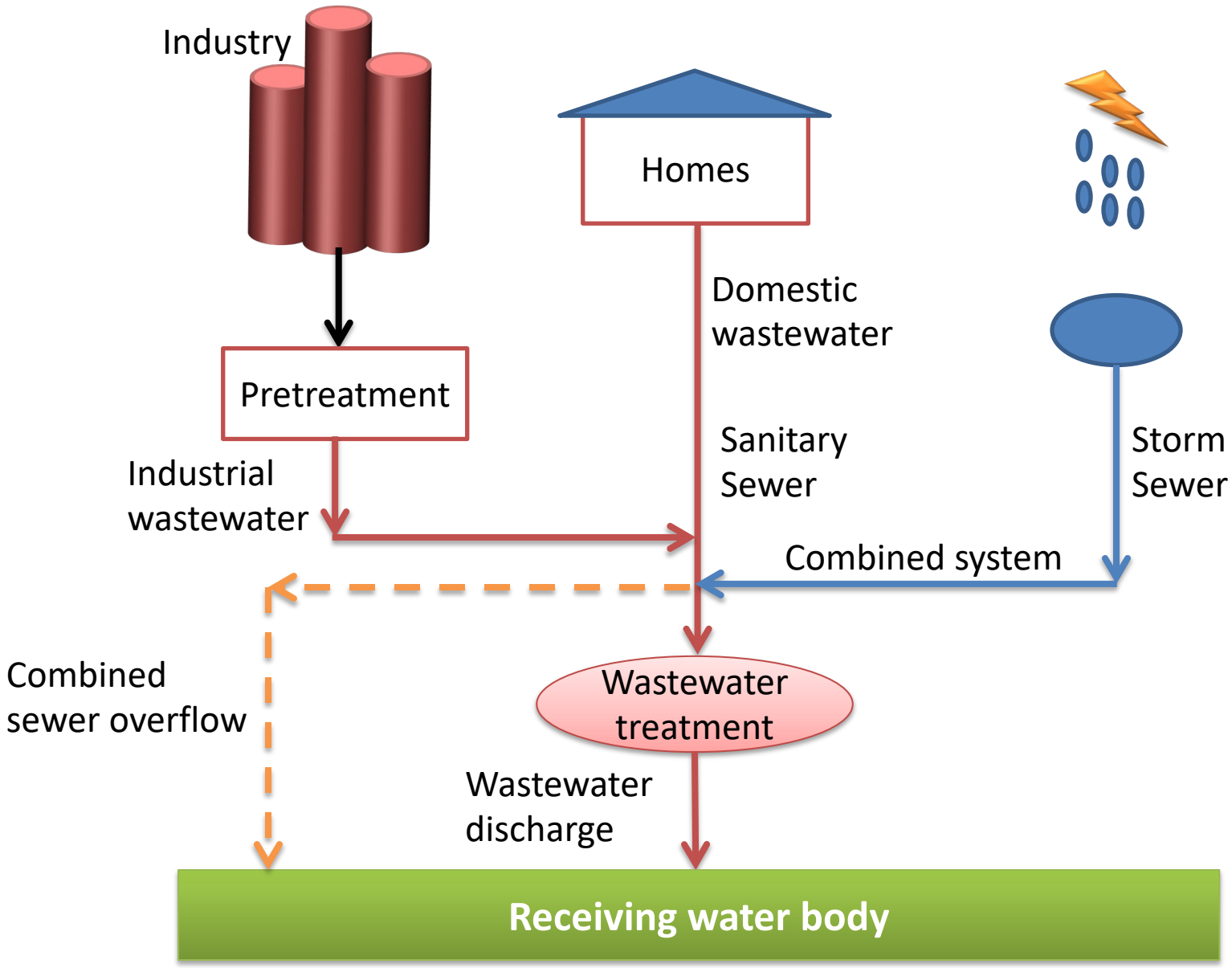


Wastewater system



- Purposes
  - Kill pathogens
  - Removal of toxins
  - Reduce BOD and nutrients

Wastewater system



Safe drinking water act (SDWA)

- Legislation in USA to protect drinking water quality
  - passed in 1974, 1) requires USEPA to establish national standards for drinking water quality, 2) requires drinking water providers to monitor/treat water to meet the drinking water standards
  - 1986 amendments: 1) quicken the pace of standards setting; 2) control lead (Pd) in water distribution systems
  - 1996 amendments: 1) requires drinking water providers to report drinking water quality data to their customers annually, 2) additional control on harmful pollutants (e.g., Cryptosporidium)
  - 2002: address water system security for concerns that contaminants may be intentionally put into drinking water system

USEPA – United States Environment Protection Agency

Safe drinking water act (SDWA)

- Drinking water standards
  - Primary standards
    - legally enforceable, based on health related issues
    - Balance health benefits vs. technological/economical feasibility
  - Secondary standards
    - Non-enforceable,
    - Aesthetic considerations: color, odor, taste, …
    - Non-aesthetic considerations: corrosivity and hardness

Safe drinking water act (SDWA)

- Primary standards (Table 6.1)
  - Maximum Contamination Level Goal (MCLG), ideal level of each contaminant with no known or anticipated health effects (not enforceable)
  - Maximum Contamination Level (MCL), legally enforceable level of each contaminant after considering balancing benefits against technological/economical feasibility
  - Treatment technique (TT). For contaminants that are difficult or time-consuming to detect/measure, a TT may specified instead of MCL.
    - Inorganic contaminants
    - Organic contaminants
    - Disinfection by-products (DBPs)
    - Radionuclides
    - Microbial contaminants

Chemical standards

Safe drinking water act (SDWA)

TABLE 6.1

Maximum Contaminant Levels (mg/L) for Drinking Water Organized by Category

Inorganic Contaminants			
Antimony	0.006	Cyanide (as free cyanide)	0.2
Arsenic	0.010	Fluoride	4.0
Asbestos (fiber length > 10 μm)	7 MFL[a]	Lead	TT[b]
Barium	2	Mercury (inorganic only)	0.002
Beryllium	0.004	Nitrate (as N)	10
Cadmium	0.005	Nitrite (as N)	1
Chromium (total)	0.1	Selenium	0.05
Copper	TT[b]	Thallium	0.002
Organic Contaminants			
Acrylamide	TT[b]	Epichlorohydrin	TT[b]
Alachlor	0.002	Ethylbenzene	0.7
Atrazine	0.003	Ethylene dibromide (EDB)	0.00005
Benzene	0.005	Glyphosate	0.7
Benzo(a)pyrene	0.0002	Heptachlor	0.0004
Carbofuran	0.04	Heptachlor epoxide	0.0002
Carbon tetrachloride	0.005	Hexachlorobenzene	0.001
Chlordane	0.002	Hexachlorocyclopentadiene (HEX)	0.05
Chlorobenzene	0.1	Lindane	0.0002
2,4-D	0.07	Methoxychlor	0.04
Dalapon	0.2	Oxamyl (Vydate)	0.2
1,2-Dibromo-3-chloropropane (DBCP)	0.0002	Pentachlorophenol	0.001
o-Dichlorobenzene	0.6	Picloram	0.5

More listed contaminants…… (textbook, p285-286)

Safe drinking water act (SDWA)

TABLE 6.1 (continued)

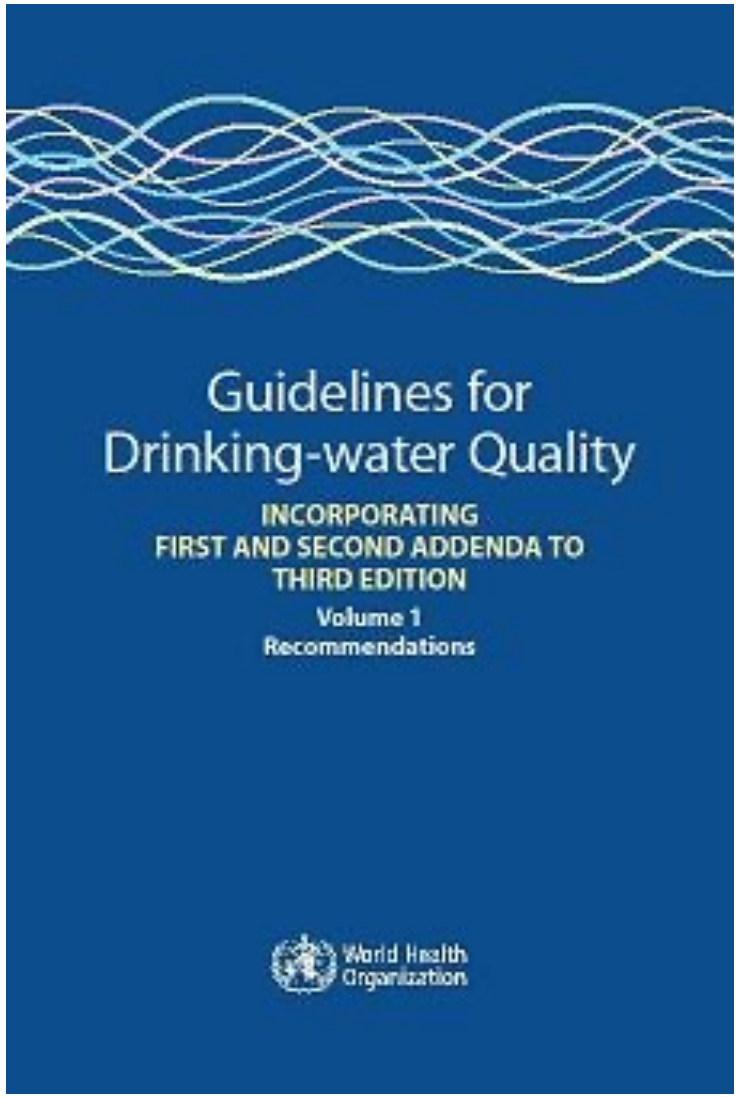
Maximum Contaminant Levels (mg/L) for Drinking Water Organized by Category

Radionuclides			
Combined Radium-226 and Radium-228	5 pCi/L	Beta particles and photon emitters	4 mrem/yr
Gross alpha emitters[c]	15 pCi/L	Uranium	0.030
Microbiological Contaminants			
Cryptosporidium	TT[b]	Total coliforms (fecal coliform & *E. coli*)	5.%[d]
Giardia lamblia	TT[b]	Turbidity	TT[b]
Heterotrophic plate count (HPC)	TT[b]	Enteric viruses	TT[b]
Legionella	TT[b]		
Disinfectants/Disinfectant By-Products			
Bromate	0.010	Chlorite	1.0
Chloramines (as Cl_2)	4.0[e]	Haloacetic acids (HAA5)	0.060
Chlorine (as Cl_2)	4.0[e]	Total trihalomethanes (TTHMs)	0.080
Chlorine dioxide (as ClO_2)	0.8[e]		

[a]7 million fibers/liter.
[b]Treatment technique (TT) requirement rather than an MCL.
[c]Excludes radon and uranium contributions.
[d]No more than 5% of samples test positive for coliforms in a month.
[e]Maximum residual disinfectant level—highest concentration of a disinfectant allowed in drinking water.

Treatment techniques (TTs) for microbiological contaminants (e.g., membrane filtration): 4-log removal for viruses, 3-log removal Giardia, and 2-log removal for Crypotosporidium

WHO guidelines



1. Introduction

1.1 General considerations and principles

- 1.1.1 Microbial aspects → biological
- 1.1.2 Disinfection → biological
- 1.1.3 Chemical aspects → chemical
- 1.1.4 Radiological aspects → radionuclides
- 1.1.5 Acceptability aspects → aesthetic

Table A4.3 Guideline values for chemicals that are of health significance in drinking-water

Chemical	Guideline value[a] (mg/litre)	Remarks
Acrylamide	0.0005[b]	
Alachlor	0.02[b]	
Aldicarb	0.01	Applies to aldicarb sulfoxide and aldicarb sulfone
Aldrin and dieldrin	0.00003	For combined aldrin plus dieldrin
Antimony	0.02	
Arsenic	0.01 (P)	
Atrazine	0.002	
Barium	0.7	
Benzene	0.01[b]	
Benzo[*a*]pyrene	0.0007[b]	
Boron	0.5 (T)	
Bromate	0.01[b] (A, T)	

...
Refer to Table A4.3 for full details

http://www.who.int/water_sanitation_health/dwq/gdwq3rev/en/

Water treatment and distribution in HK

Existing water resources



Rainwater 21%

Imported 55%

Seawater 24%

~ 1.3 billion m^3 in 2019

New water resources



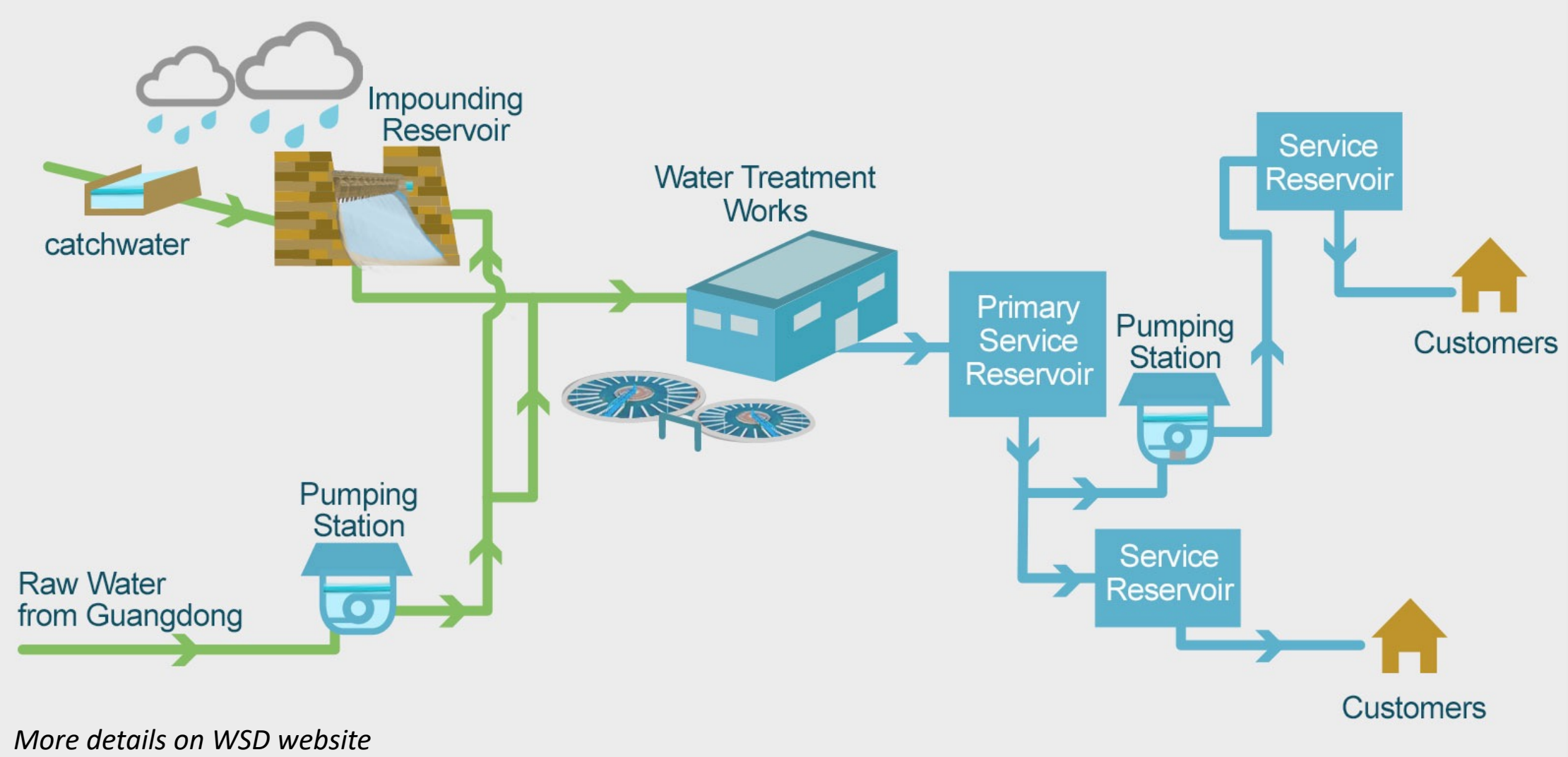
Desalination

Recycled water

1. Tseung Kwan O Desalination Plant, up to 270,000 m^3/day, **5~10%** fresh water demand (expected).
2. Recycled water for toilet flushing and other non-potable use.

Data provided by WSD

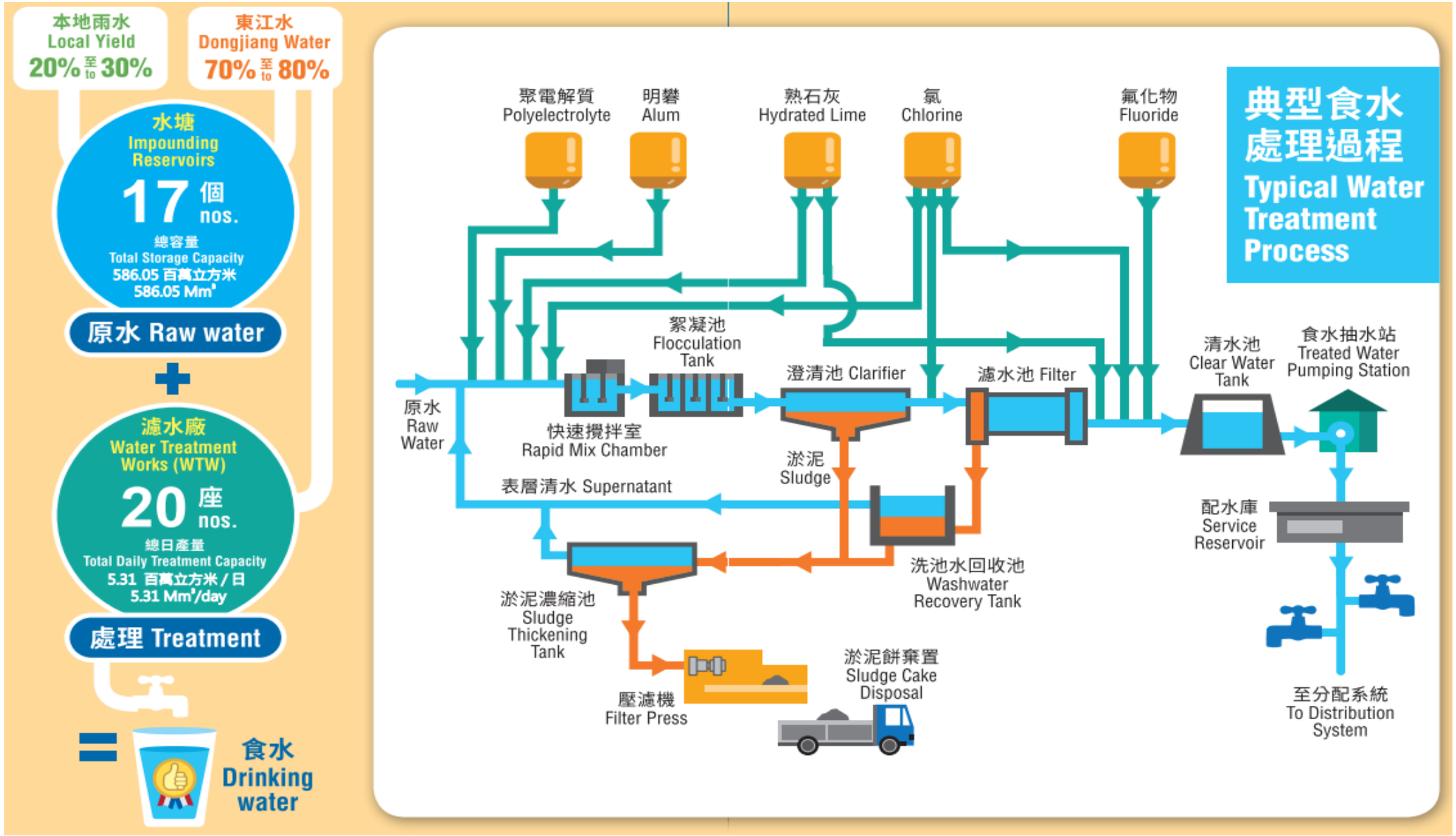
Water treatment and distribution in HK



More details on WSD website

https://www.wsd.gov.hk/en/core-businesses/operation-and-maintenance-of-waterworks/index.html

Water treatment and distribution in HK



https://www.wsd.gov.hk/filemanager/en/share/pdf/water_treat.pdf

Water treatment

- Treatment targets:
  - Pathogen free
  - chemically safe
  - aesthetically acceptable, free from turbidity, color, odor, and objectionable taste
- Surface water
  - High in suspended solids (particles)
  - Susceptible to biological contamination
  - particle (including biological particles) removal , disinfection
- Groundwater
  - Low in suspended solids content
  - High in dissolved solutes (Ca, Mg, Fe, Mn, …)
  - May contain volatile organic compounds
  - Softening, disinfection

Surface water treatment systems

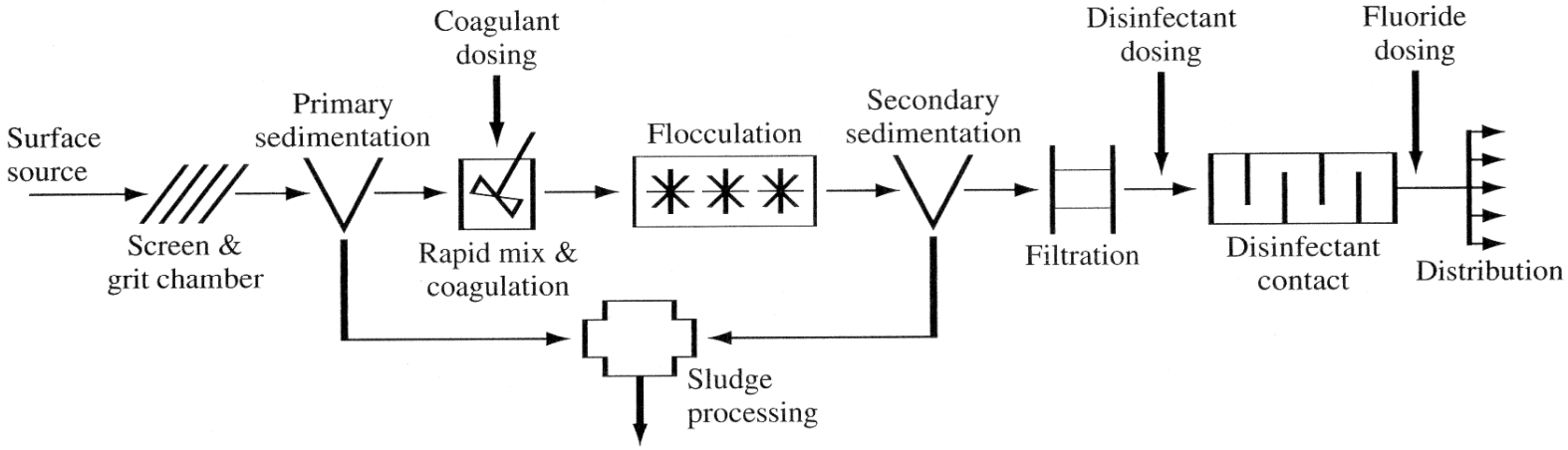


FIGURE 6.2 Schematic of a typical surface water treatment plant.

- **Screening and grit removal** – large debris and particles removal
- **Primary/secondary sedimentation** - gravity settling of particles, also called clarification
- **Rapid mixing and coagulation** – use chemicals (coagulant, e.g., alum) to destabilize fine particles to form larger particles under agitation
- **Flocculation** – gently mixing water to form large flocs that are more easily settle. It may requires the addition of flocculants
- **Filtration** – removal of fine particles
- **Sludge processing** – dewatering and disposal of solids and liquids
- **Disinfection** – add disinfectant (e.g., chlorine) to inactivate pathogens

Groundwater treatment systems

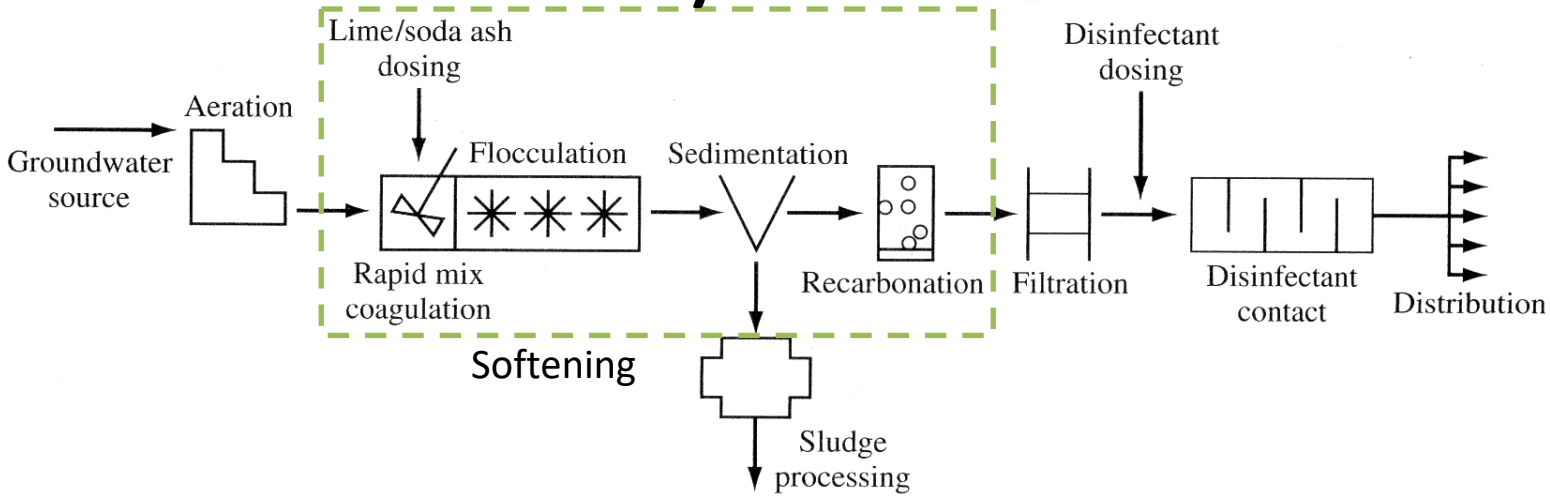


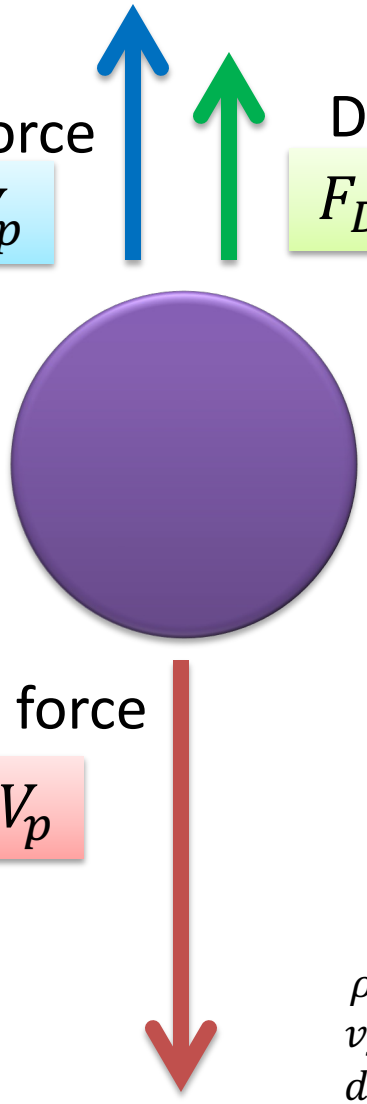
FIGURE 6.3 Schematic of a typical water treatment plant for groundwater, including softening for calcium and magnesium removal.

- **Aeration** – removes gases (may also cause precipitation of Fe and Mn)
- **Lime/soda ash dosing, rapid mixing** – reduce solubility of Ca and Mg
- **Flocculation** – gently mixing water to form large flocs
- **Sedimentation** - gravity settling of particles/flocs
- **Recarbonation** – adjust water pH and alkalinity
- **Filtration** – removal of fine particles
- **Sludge processing** – dewatering and disposal of solids and liquids
- **Disinfection** – add disinfectant (e.g., chlorine) to inactivate pathogens

Table 5.1 Water treatment unit processes.

Process	Purpose
Preliminary Treatment	
Screening	Remove large debris
Microstraining	Remove algae, aquatic plants and small debris
Chemical pretreatment	Condition the water for removal of algae, microorganisms, organics.
Pre-sedimentation	Remove gravel, sand, silt and other gritty material
Aeration	Remove odours and dissolved gases, add oxygen to improve taste
Main Processes	
Coagulation/flocculation	Convert non-settleable particles to settleable particles
Softening	Remove hardness in water
Sedimentation	Remove settleable particles
Filtration	Remove suspended flocs and most microorganisms
Adsorption	Remove organics and colour
Stabilisation	Prevent scaling and corrosion
Fluoridation	Add fluoride in order to harden tooth enamel
Disinfection	Kill disease-causing organisms

Sedimentation



Buoyancy force

$F_B = \rho g V_p$

Drag force

$F_D = 3\pi\mu v_s d_p$

Gravitational force

$F_G = \rho_p g V_p$

Settling velocity (Steady)

$$F_D = F_G - F_B$$

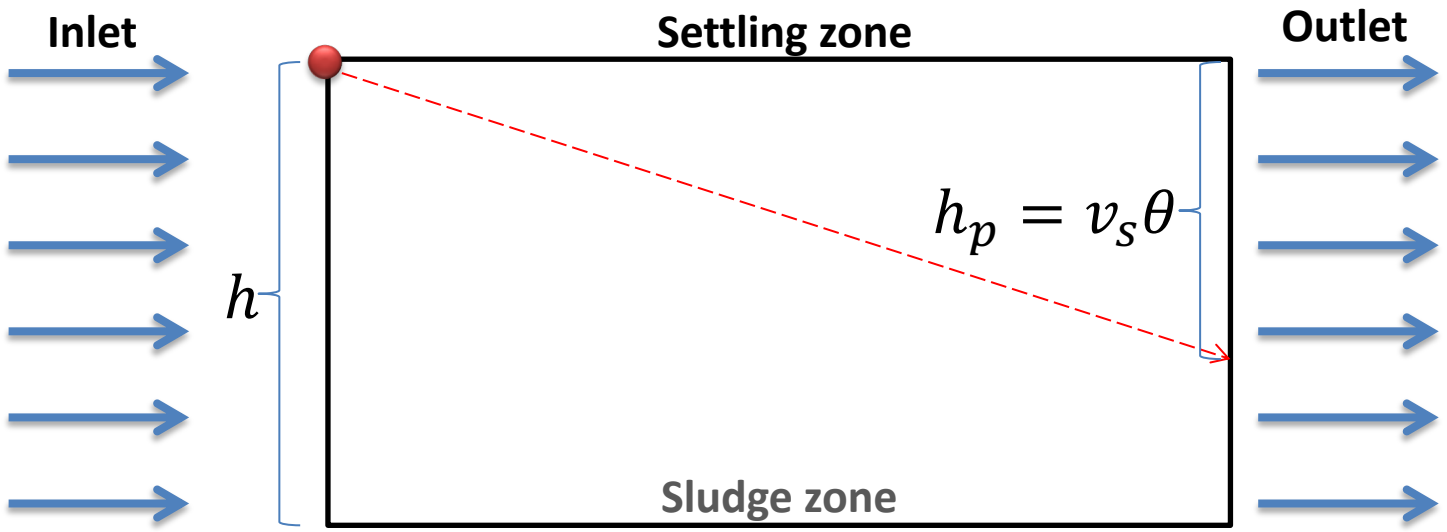
$$3\pi\mu v_s d_p = \rho_p g \frac{\pi {d_p}^3}{6} - \rho g \frac{\pi {d_p}^3}{6}$$

$$v_s = \frac{(\rho_p - \rho) g {d_p}^2}{18\mu}$$

$$v_s = \frac{(s_p - 1) \rho g {d_p}^2}{18\mu}$$

ρ: *density of water*
v_s: particle settling velocity
d_p: particle hydrodynamic diameter
μ: *absolute viscocity of water*
g: gravitational acceleration (9.8 m/s^2)

Sedimentation



Critical sedimentation velocity v_0

$v_0\theta = h$ where hydraulic retention time (HRT) $\theta = \frac{V_{reactor}}{Q}$

$v_0 = \frac{h}{\theta} = \frac{h}{V_r/Q} = \frac{Q}{A_r}$ (also termed surface overflow rate)

and removal efficiency

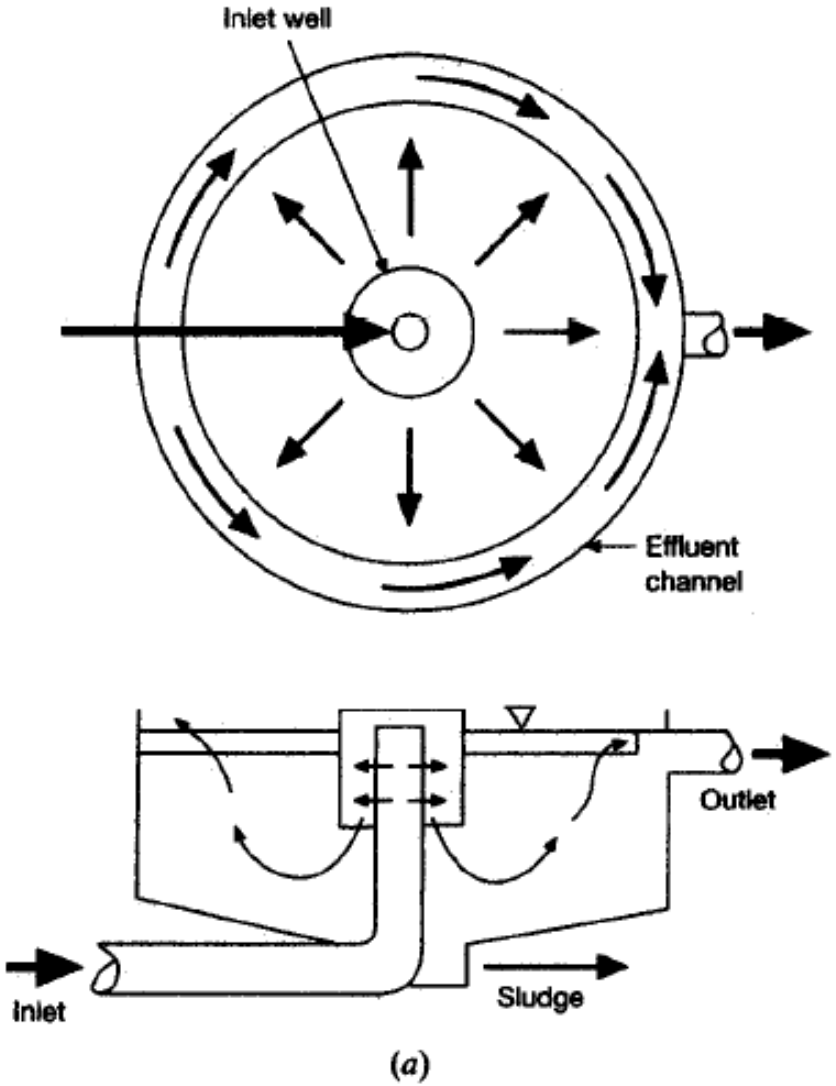
$v_0 = \frac{Q}{A_r}$ $A_r = \frac{18\mu Q}{(\rho_p - \rho)g{d_p}^2}$ $\eta = \frac{v_p}{v_0}$

Note:
Typical HRT for sedimentation tank 1 – 4 hours
Typical surface overflow rate 1 – 2.5 m/h

Sedimentation



http://lgam.wikidot.com/sedimentation-tank

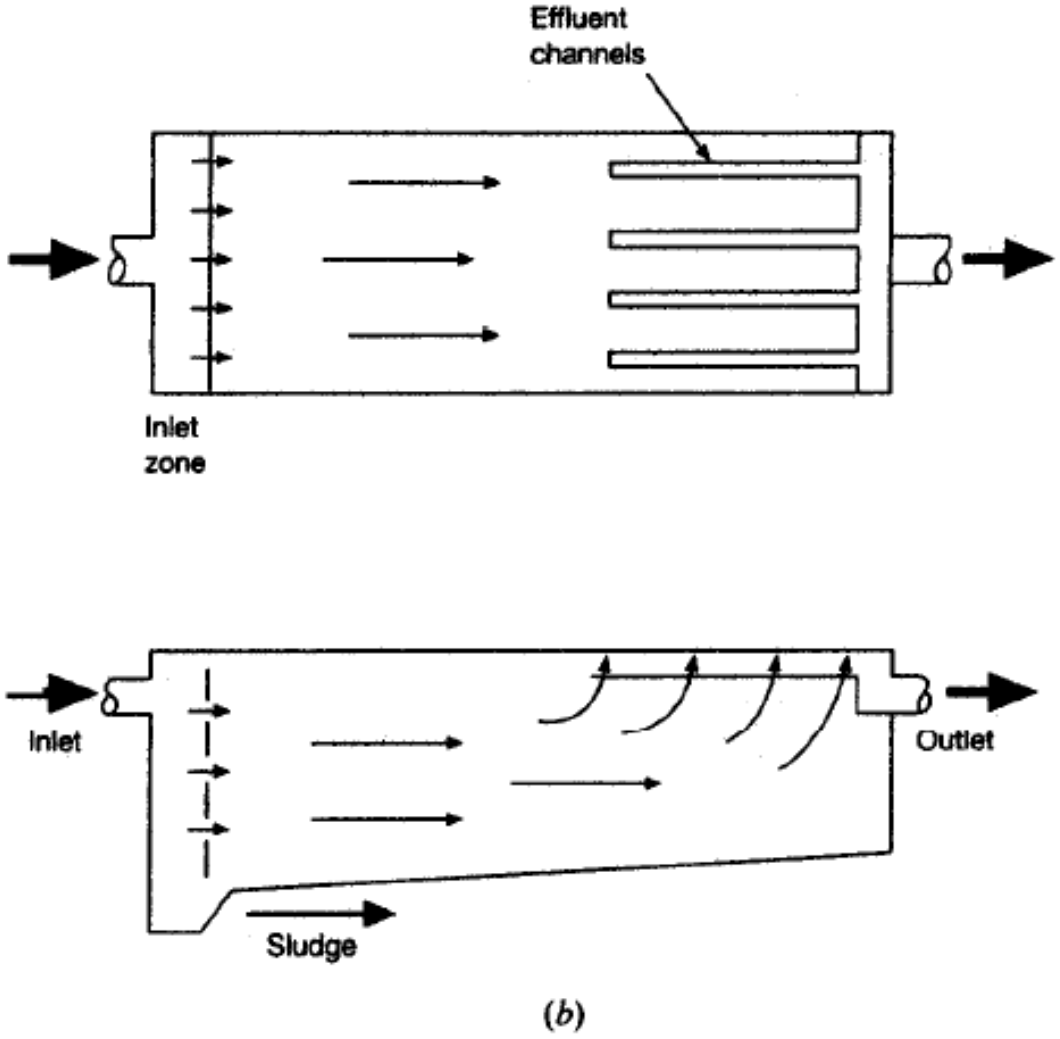


Sedimentation



Horizontal sedimentation tank

http://www.lagoswater.org/photo_gallery.php



Example 6.1 Silt removal in a clarifier

A drinking water treatment plant uses a circular sedimentation basin to treat 3.0 MGD of river water (MGD stands for million gallons per day, 1 MGD = 0.0438 m^3/s). After storms occur upstream, the river often carries 0.010 mm silt particles with an average density of 2.2 g/cm^3, and the silt must be removed before the water can be used. The plant's clarifier is 3.5 m deep and 21 m in diameter. The water is 15 °C.

(a) What is the hydraulic retention time of the clarifier?

(b) Will the clarifier remove all of the silt particles from the river water?

Solutions:

(a) The hydraulic retention time (i.e., θ) of a flow-through vessel is

$$\theta = V/Q$$

where V is the vessel's volume, and Q is the flow rate through the vessel. The volume of a circular basin is:

$$V = \frac{\pi d_b^2 h}{4}$$

where d_b is the basin's diameter, and h is depth. So, the hydraulic retention time is:

$$\theta = \frac{\pi d_b^2 h}{4Q} = \frac{3.14\times 21^2 \times 3.5}{4\times 3.0\times 0.0438} = 9226\ \text{s} = \mathbf{2.6\ hr}$$

(b) To find if the silt particles will be removed, we need to calculate their settling velocity and compare it to the critical velocity for the sedimentation basin. We first need the water density ρ (999.1 kg/m^3 at 15 °C) and the viscosity μ (0.00114 kg/m/s). According to Stokes' law:

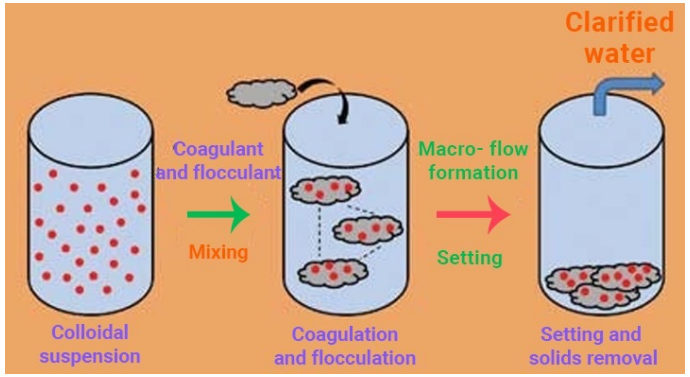
$$v_s = \frac{g(\rho_p - \rho)d_p^2}{18\mu} = \frac{9.807\times(2200 - 999.1)\times 0.00001^2}{18\times 0.00114} = 5.74\times 10^{-5}\ \text{m/s}$$

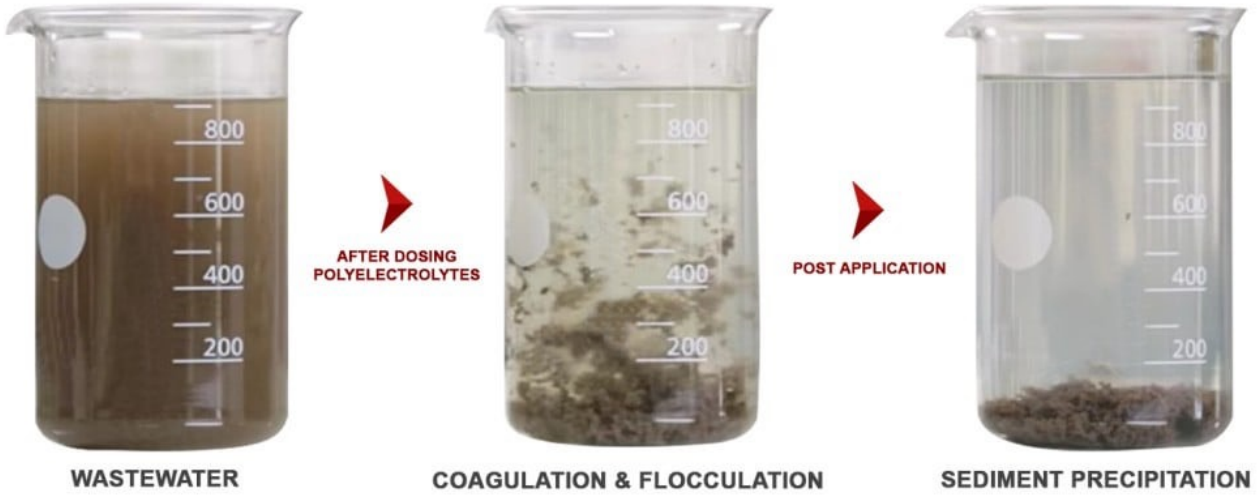
The critical velocity for the sedimentation basin is:

$$v_0 = \frac{h}{\theta} = \frac{3.5}{9226} = 38\times10^{-5}\ \mathrm{m/s}$$

v_s is much less than v_0, so the silt will not be removed completely by the clarifier.

Coagulation and flocculation





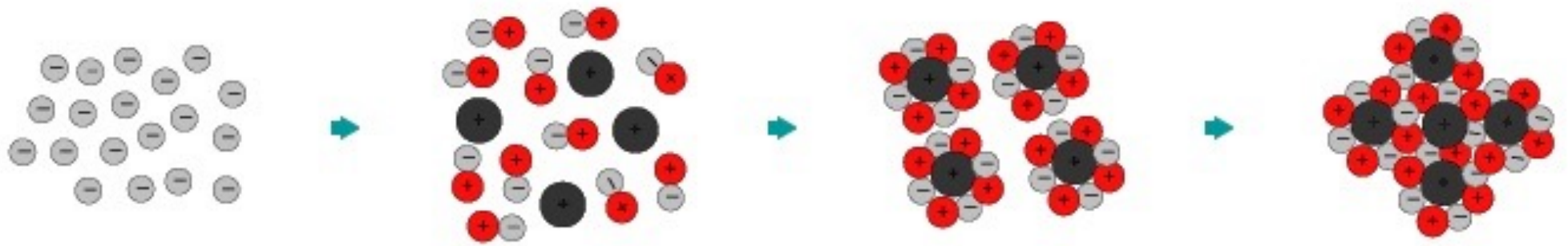
https://www.toppr.com/ask/content/story/amp/properties-of-colloidal-solutions-v-73079/

Coagulation-flocculation is a chemical water treatment technique typically applied prior to sedimentation and filtration (e.g. rapid sand filtration) to enhance the ability of a treatment process to remove particles. **Coagulation** is a process used to neutralize charges and form a gelatinous mass to trap (or bridge) particles thus forming a mass large enough to settle or be trapped in the filter. **Flocculation** is gentle stirring or agitation to encourage the particles thus formed to agglomerate into masses large enough to settle or be filtered from solution.

Author/Compiled by Félicien Mazille & Dorothee Spuhler

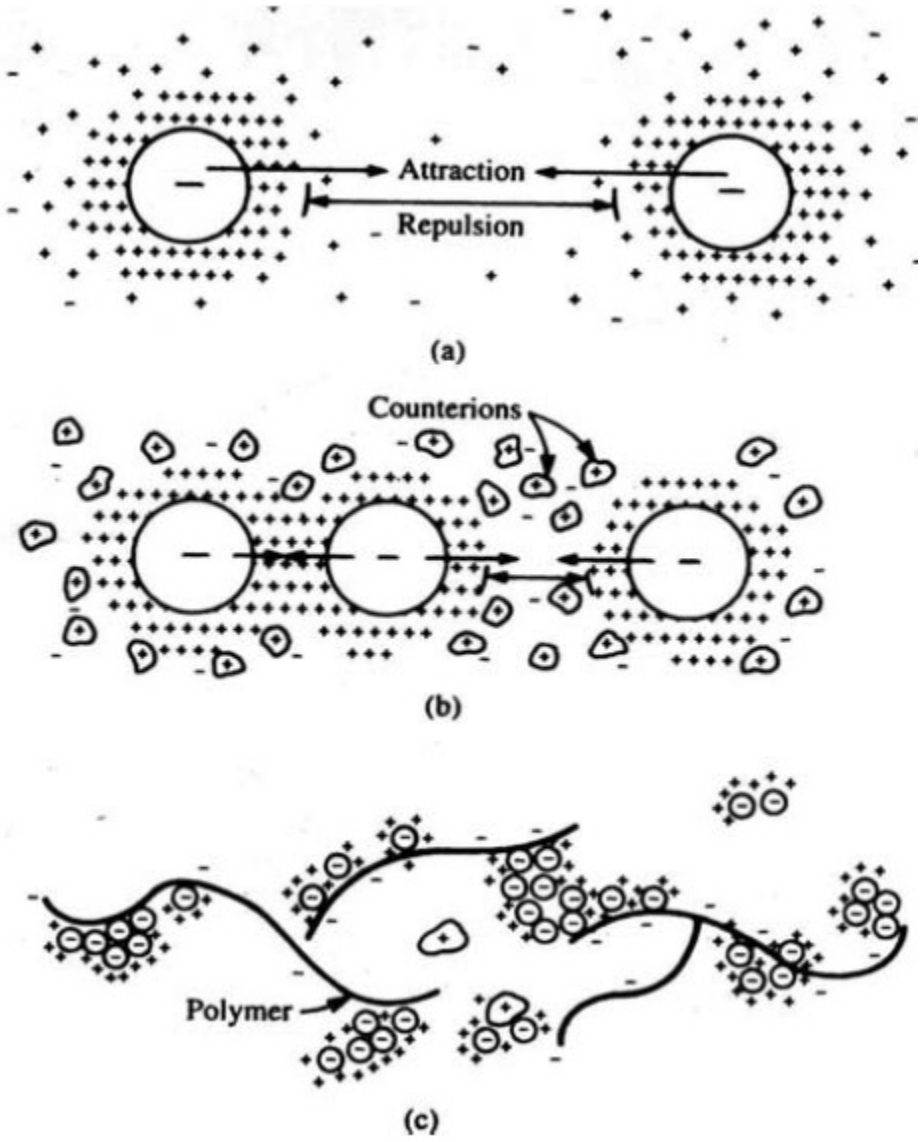
Coagulation and flocculation

- Colloidal particles
  - fine particles, ~ 1 nm – 1 μm in size → difficult to settle
  - Natural colloids in surface water are typically negatively charged
  - Electric repulsive force → stability
- Colloids removal by coagulation and flocculation
  - Charge neutralization and compression of electric repulsion
  - Trapping (entrapment)
  - Inter-particle bridging



https://nihonkasetsu.com/the-mechanism-of-coagulation-and-flocculation/

Coagulation and flocculation



Destabilization of colloids

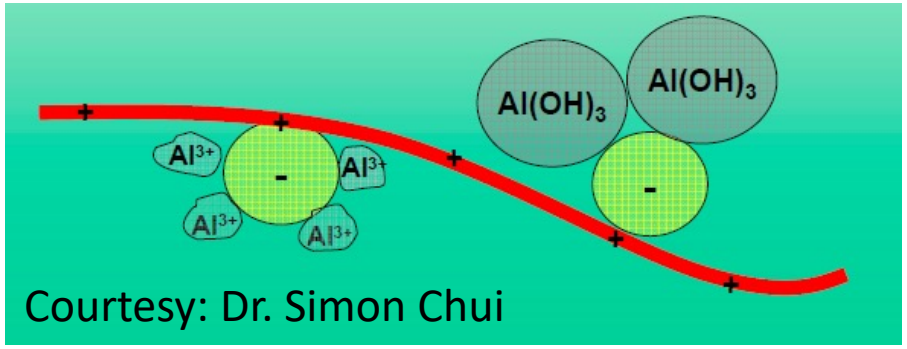
(a) stable suspension of particles where repulsive force greater than attraction force

(b) addition of coagulant suppressing double layer charge

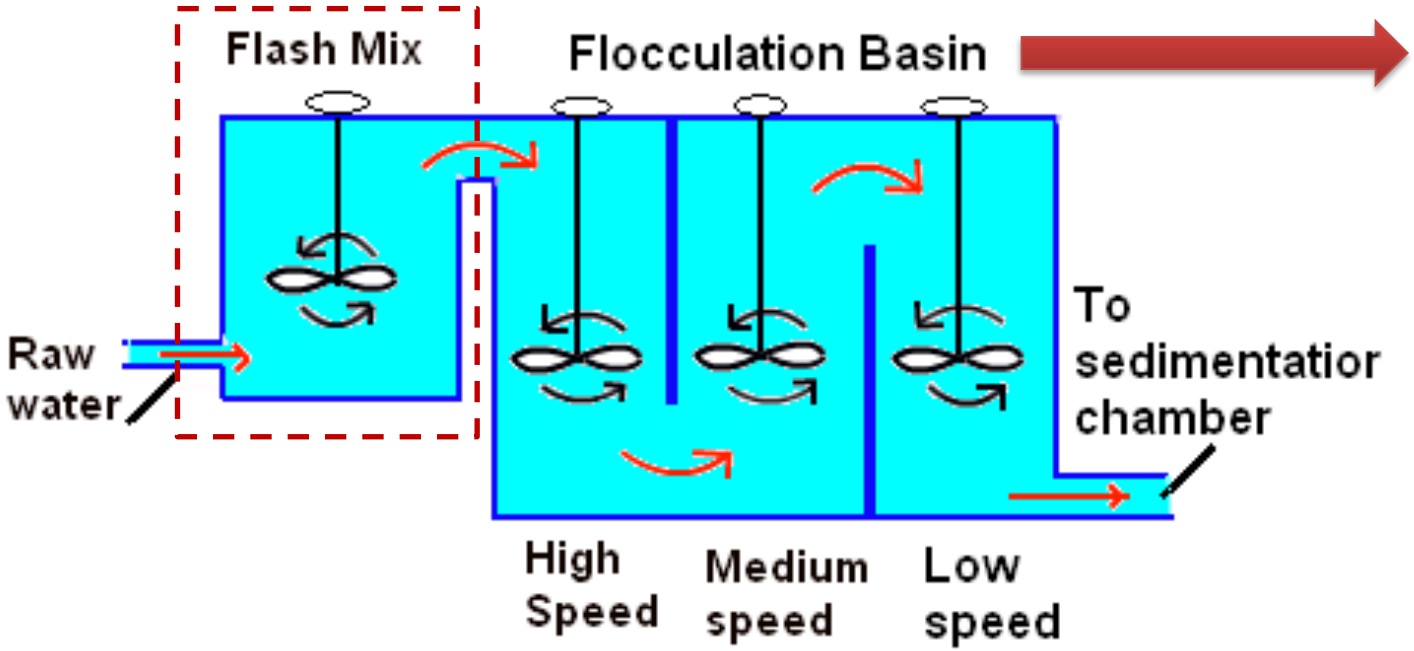
(c) agglomeration of destabilized particles by coagulant and polymer bridging.

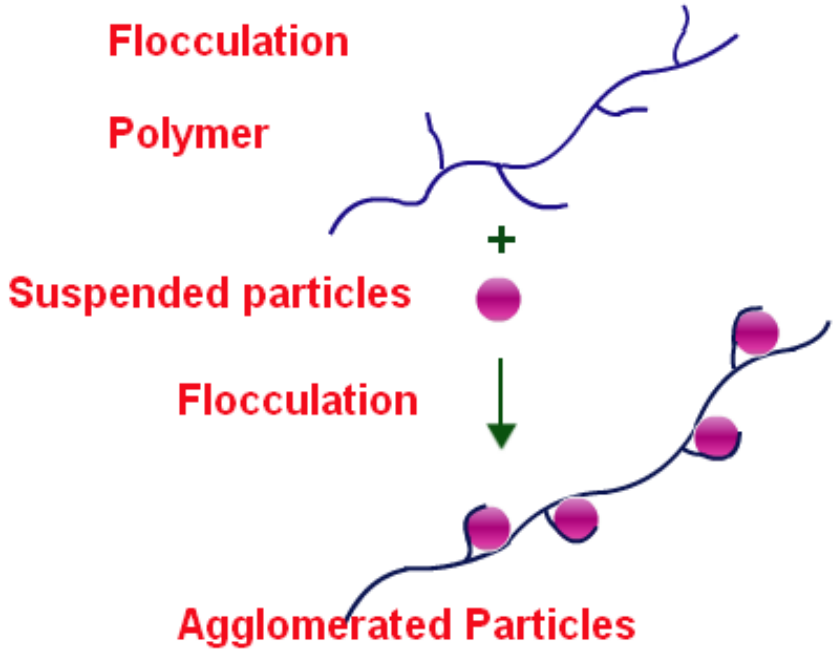
Source: Viessman & Hammer

Coagulation and flocculation



Coagulation





http://chemistry.tutorvista.com/physical-chemistry/flocculation.html

Coagulation

- Chemical process, forming insoluble precipitates
- Common coagulants
  - Aluminum sulfate (alum): **$Al_2(SO_4)_3 \cdot 14H_2O$**
  - Others: $FeCl_3$, $FeSO_4$

$$\mathbf{Al_2(SO_4)_3 \cdot 14H_2O + 6HCO_3^- \rightarrow 2Al(OH)_3\downarrow + 6CO_2 + 14H_2O + 3SO_4^{2-}}$$

MW = 594

- The reaction consumes HCO_3^- (bicarbonate alkalinity)
- For source water without sufficient alkalinity, the reaction reduces pH and affect the formation of insoluble aluminum hydroxide
- $Ca(OH)_2$ or Na_2CO_3 may be added to supply the required alkalinity

Flocculation

- Mixing intensity *G*
- Gentle mixing required
  - To maximize contact of destabilized particles (microflocs) and to form large flocs
  - To avoid excessive breakage of flocs

$$G = \sqrt{\frac{P}{\mu V_r}}$$

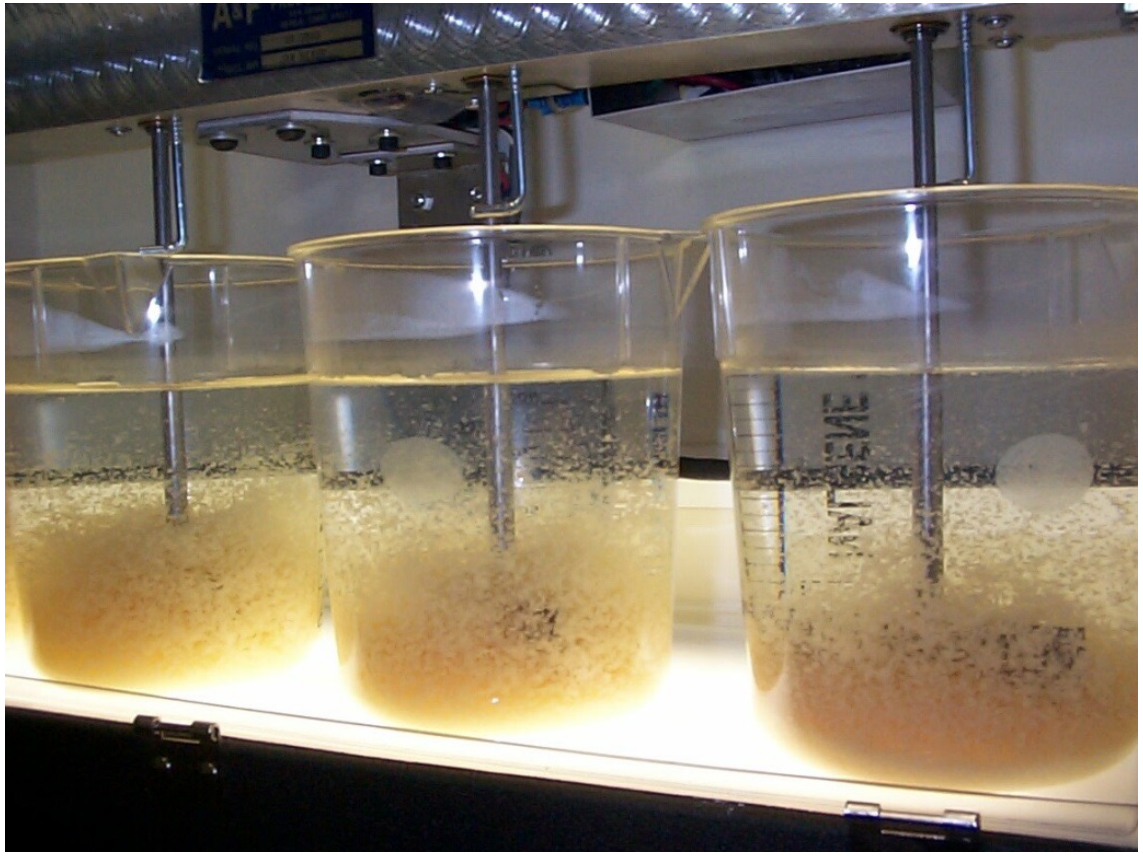
- P: Power imparted to the water, Nm/s or W.
- V_r: Volume of water in tank, m^3.
- μ: Absolute viscosity, Ns/m^2 or kg/ms.

Type of water	G (1/s)	Gt (-)
Low turbidity	20-70	60,000-200,000
High turbidity	50-150	90,000-180,000

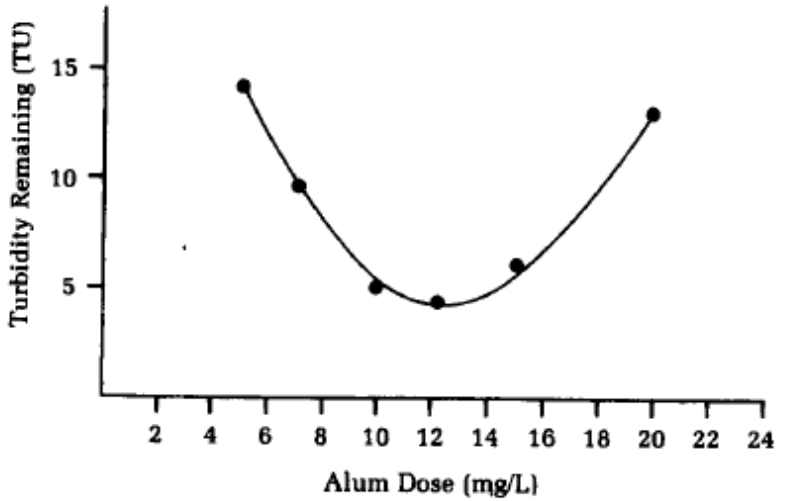
Optimal condition: G = 20 – 70 S^{-1}, t > 30 min

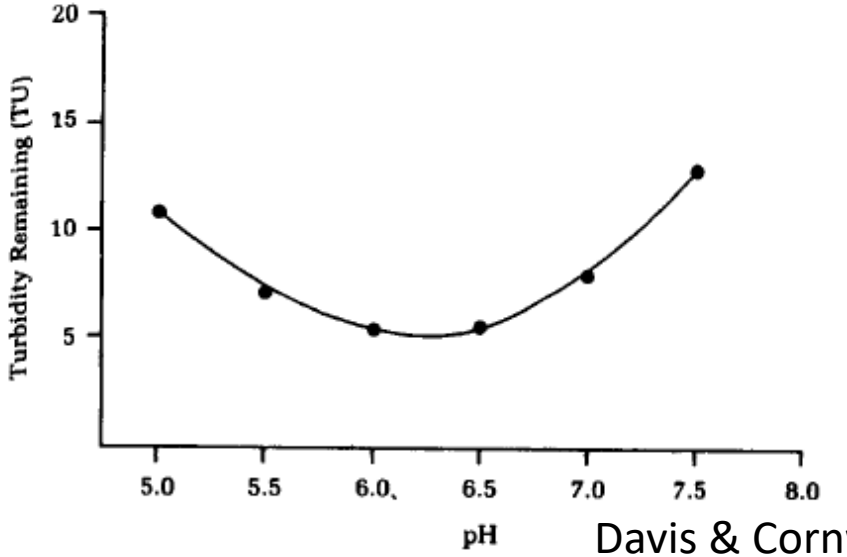
Jar test

Jar test



http://www.qgsquimica.com.br/qgs/det_comunicacao.php?id=57&det_comunic=3





Davis & Cornwell, 1998

YouTube video: http://www.youtube.com/watch?feature=player_detailpage&v=OWvM33L0-j4

Filtration

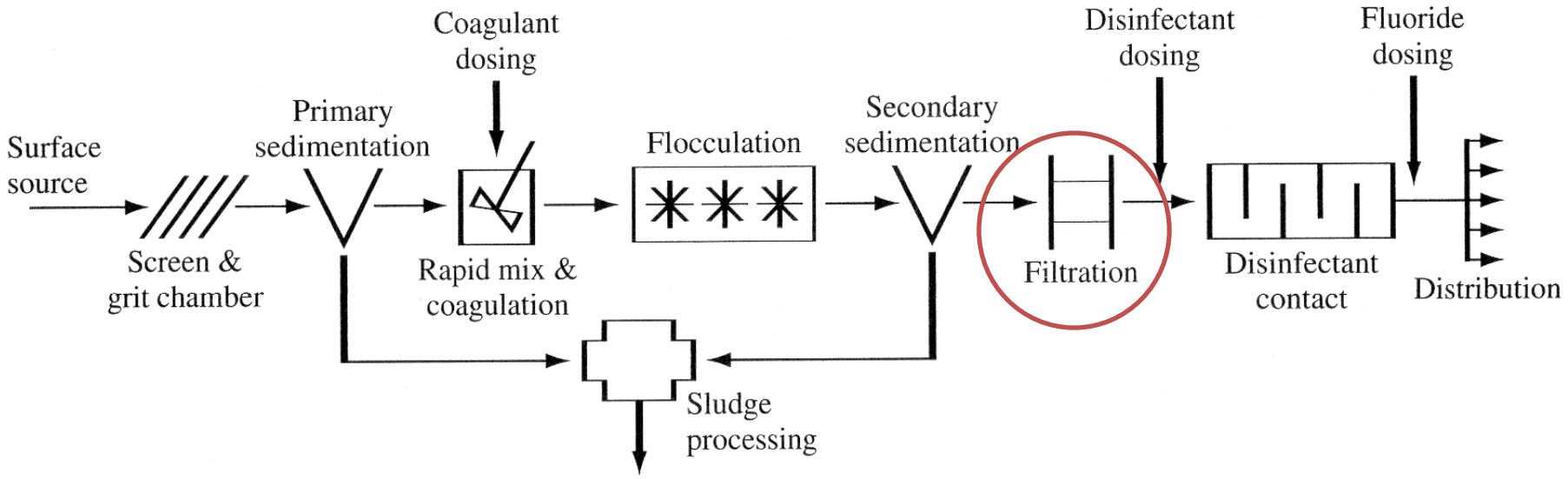
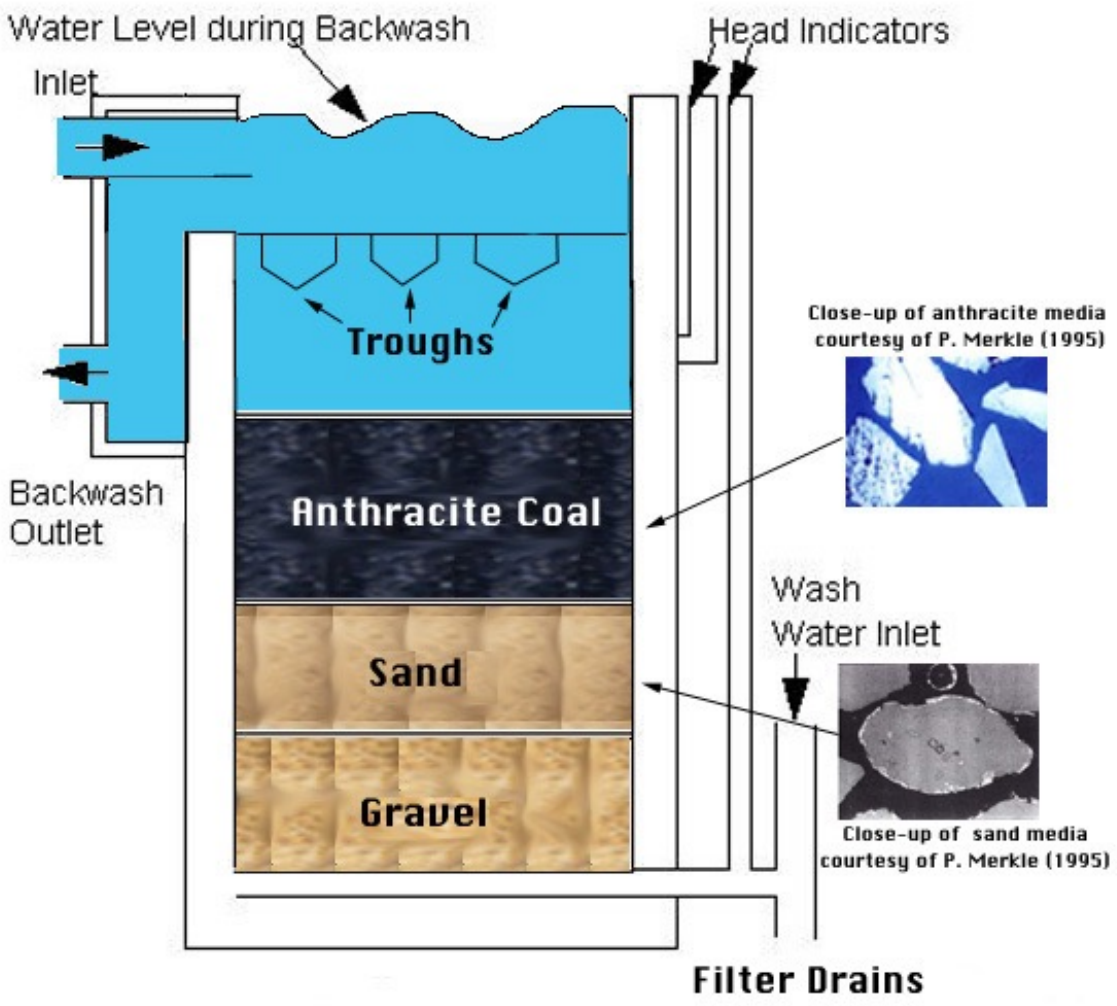


FIGURE 6.2 Schematic of a typical surface water treatment plant.

- To remove small particles (including pathogens)
- Rapid depth filtration
  - Sand, anthracite coal, …
  - Pore opening > particle size
  - Backwashing required when filter is clogged

Filtration



Filtration rates

Also termed loading rates or superficial velocities

$$v_a = \frac{Q}{A_f}$$

Typical filtration rates: 5 – 25 m/hr

A_f: filter's cross-sectional area

http://www.webapps.cee.vt.edu/ewr/environmental/teach/wtprimer/rapid/rapid.html

Filtration

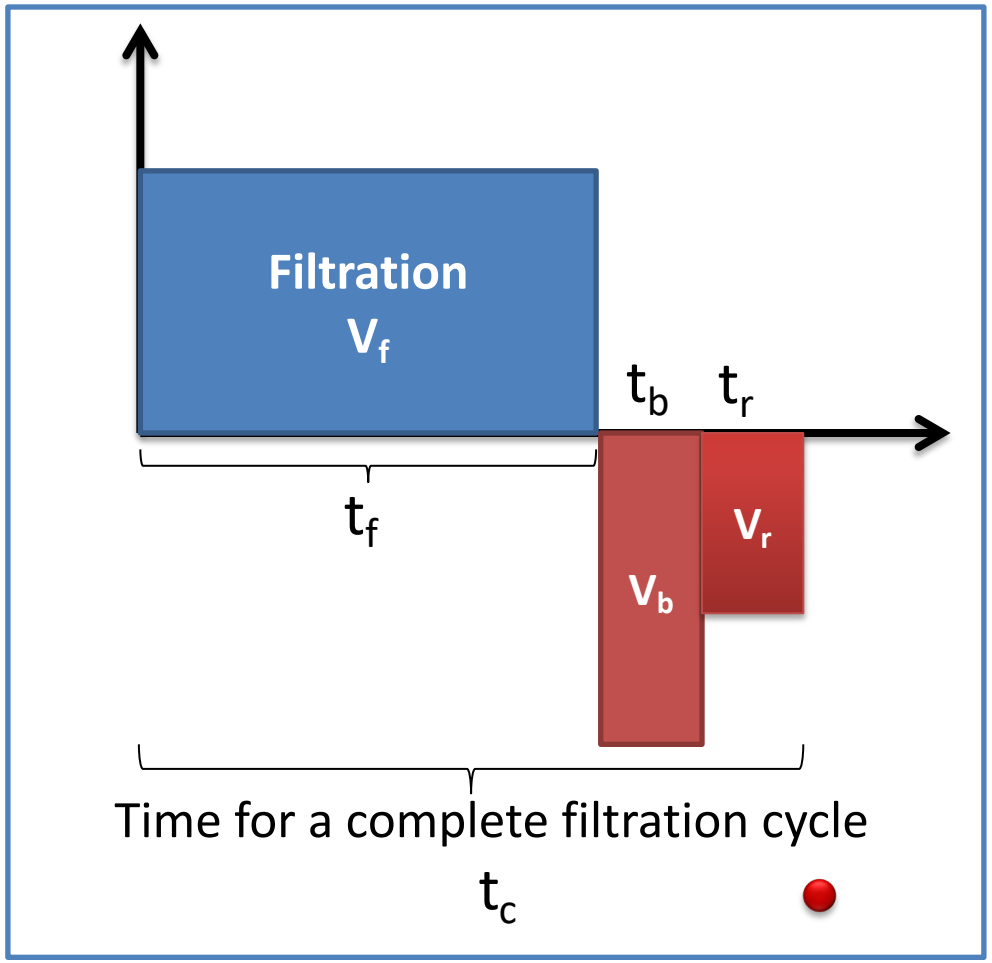
- Back flushing and rinsing are needed to clean clogged filters
- Filter efficiency

$$\eta_f = \frac{V_f - V_b - V_r}{V_f}$$

- Effective filtration rate

$$v_{eff} = \frac{V_f - V_b - V_r}{A_f t_c}$$

V_f: volume of water filtered per filter cycle (m3)
V_b: volume of water used to backwash the filter (m3)
V_r: volume of water used to rinse the filter after backwash (m3)



Disinfection

- Objectives
  - Primary disinfection: to kill pathogens in water
  - Secondary disinfection: to prevent pathogen regrowth in water (in water distribution systems)
- Disinfectants
  - Free chlorine (the most common disinfectant)
    - Chlorine gas ($Cl_{2(g)}$)
    - Hypochlorite salts (NaOCl or $Ca(OCl)_2$)
  - Alternative disinfectants
    - Chlorine dioxide (ClO_2) ← Expensive, explosive
    - Ozone (O_3)
    - UV

] More expensive, No protective residual

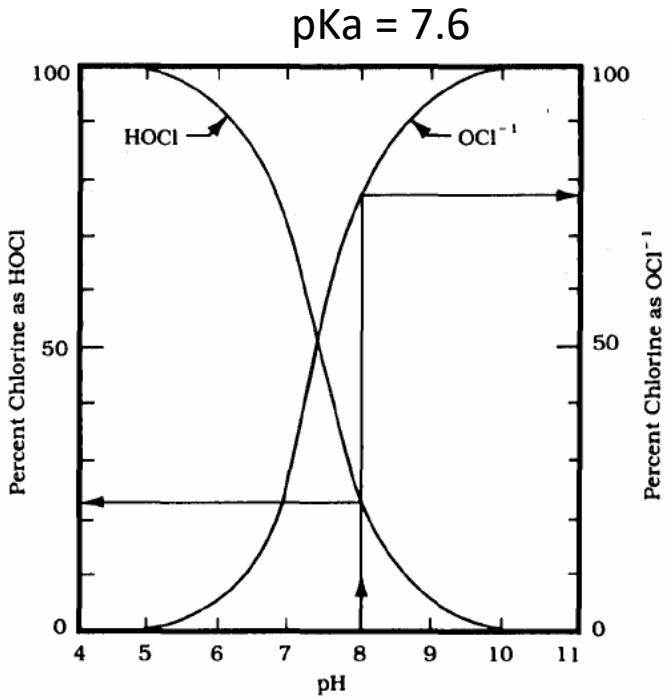
Disinfection

- Chlorination

$Cl_{2(g)} + H_2O = HOCl + H^+ + Cl^-$

$HOCl = H^+ + OCl^-$

Disinfection power: $HOCl >> OCl^-$
→ disinfection at slightly acidic pH



– Effective against bacteria,
– Less effective for protozoan cysts (e.g., Giardia and Cryptosporidium) and viruses
– Reacts with NH_3 to form chloramines,
  - NH_2Cl, $NHCl_2$, NCl_3 (combined chlorine)
  - chloramines are weaker disinfectants but with longer residence time (good secondary disinfectant)

Disinfection

Disinfection power: ozone > chlorine dioxide > free chlorine > chloramine

Disinfectant	Inactivation (mg · min/L)					
	0.5-log	1-log	1.5-log	2-log	2.5-log	3-log
Chlorine[1]	17	35	52	69	87	104
Chloramine[2]	310	615	930	1,230	1,540	1,850
Chlorine Dioxide[3]	4	7.7	12	15	19	23
Ozone[3]	0.23	0.48	0.72	0.95	1.2	1.43

CT values were obtained from AWWA, 1991.

[1] Values are based on a free chlorine residual less than or equal to 0.4 mg/L, temperature of 10°C, and a pH of 7.

[2] Values are based on a temperature of 10°C and a pH in the range of 6 to 9.

[3] Values are based on a temperature of 10°C and a pH of 6 to 9.

Disinfection byproducts (DBPs)

- Chlorination → DBPs
  - DBPs are created when chlorine reacts with natural organic matter contained in the water
  - Trihalomethanes (THMs), e.g., $CHCl_3$ (a carcinogen)
  - Haloacetic acids (HAAs)
  - Control
    - Removal of DBPs precursors (apply chlorination after coagulation/sedimentation, improve sedimentation efficiency, membrane filtration, …)
    - Alternative disinfectants

*"The use of chemical disinfectants in water treatment usually results in the formation of chemical by-products. However, the risks to health from these by-products are extremely small in comparison with the risks associate with inadequate disinfection, and it is important that **disinfection not be compromised in attempting to control such by-products**. (IWA Guideline, 3rd Ed.)*

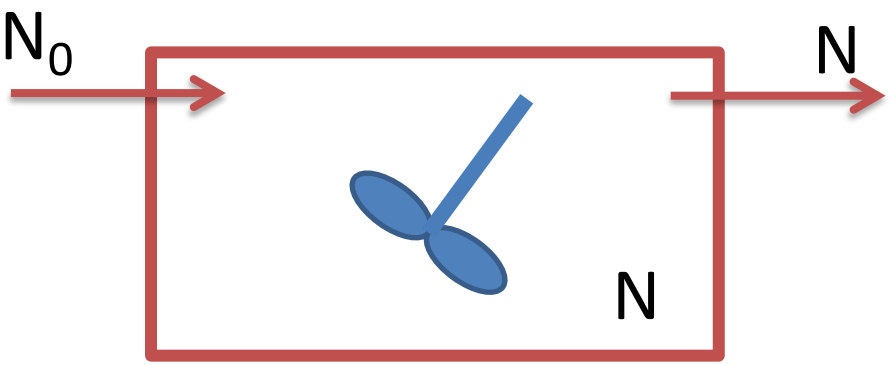
Disinfection kinetics

Chick-Watson Model: $r(N) = -kC^nN$

N: number of pathogens/unit volume
C: concentration of disinfectant
k is the coefficient of specific lethality (pathogen dependent)
n is an empirical constant (typically assume n = 1)

$$r(N) = \frac{dN}{dt} = -kCN$$

Inactivation rate constant k^*



$$N_0Q = NQ + (kCN)V$$

$$N = \frac{N_0Q}{Q + kCV}$$

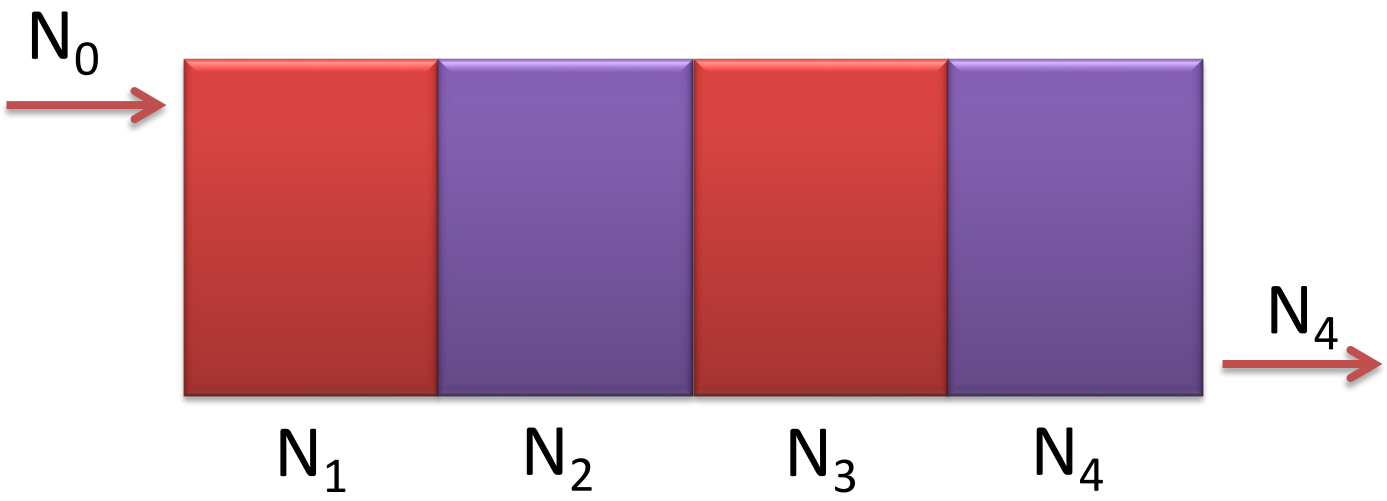
$$N = \frac{N_0}{1 + kCV/Q}$$

$$N = \frac{N_0}{1 + kC\theta}$$

θ: HRT

Example 6.3 *Giardia* inactivation by ozone

Because of the purity of its groundwater source, a water utility only has to disinfect the water before delivery. The utility uses ozone as its primary disinfectant and treats 10 MGD of water. The ozone is bubbled into the water as it passes sequentially through the four chambers of a disinfectant contact tank. Each chamber of the tank is well mixed, holds 50,000 gallons of water, an receives the same ozone dose. The US EPA requires that the *Giardia* concentration be reduced to 1/1,000 of its initial concentration (i.e., 3-log reduction). What is the minimum ozone concentration that must be used to achieve the required removal of *Giardia*? The coefficient of specific lethality and coefficient of dilution for ozone inactivation of *Giardia* under these conditions are 1.8 L/mg/min and 1.0, respectively.



Solution:

We need to do a mass balance on the tank in terms of the Giardia concentration, but the tank as a whole is not well modeled as continuous stirred tank reactor (CSTR) or a plug flow reactor (PFR). However, because each chamber is well mixed and the effluent of one chamber is the influent of the next, we can model the system as a series of CSTRs. First we develop an equation for a series of steady state CSTRs in which the volume V_c, the residence time θ_c, the flow rate Q, and the ozone concentration C, are the same in each CSTR. We make this derivation general to any series of CSTRs in which first-order decay is occurring by using Chick-Watson model as the reaction rate expression and later substituting the product of k (i.e., specific lethality coefficient) and C^n in place of k^*. According to mass balance and the disinfection model, the steady state mass balance around the first chamber is:

$$0 = QN_0 - QN_1 - Vk^*N_1$$

$$N_1 = \frac{QN_0}{Q + Vk^*}$$

As $\theta_c = V_c/Q$, and rearranging above equation as:

$$\frac{N_1}{N_0} = \frac{1}{1 + \theta_c k^*}$$

Similarly, a steady state CSTR, mass balance around the second chamber yields:

$$\frac{N_2}{N_1} = \frac{1}{1 + \theta_c k^*}, \qquad \frac{N_2}{N_0} = \frac{N_2}{N_1} \times \frac{N_1}{N_0}, \qquad \frac{N_2}{N_0} = \frac{1}{(1 + \theta_c k^*)^2}$$

Solution:

Extending this approach to *m* number of CSTRs in series gives a general equation for first-order decay in a series of steady state CSTRs as:

$$\frac{N_2}{N_m} = \frac{1}{(1+\theta_c k^*)^m}$$

For the particular case of the ozone contact tank's four chambers (m=4), the coefficient of dilution n=1, and $k^*=kC^n$, so the above equation can be rearranged as:

$$C = \frac{(\frac{N_0}{N_4})^{\frac{1}{4}} - 1}{\theta_c k}$$

Three-log removal means the effluent concentration after disinfection is 1/1,000th of the influent concentration. In other words, N_0/N_4 = 1,000. As $\theta_c=V_c/Q$, for an influent rate of 10 MGD and a contactor compartment volume of 50,000 gallons, the minimum ozone concentration required in each chamber is:

$$C = \frac{(\frac{N_0}{N_4})^{\frac{1}{4}} - 1}{\theta_c k} = \frac{1000^{1/4} - 1}{\frac{50{,}000 \times 1{,}440}{10^7} \times 1.8} = \mathbf{0.36\ mg/L}$$

Hardness

- Hardness = all multivalent metallic cations in solution

$$= \underline{\mathbf{Ca^{2+} + Mg^{2+}}} \quad (+ \ Fe^{2+} + Mn^{2+} + Sr^{2+} + Al^{3+} \ldots)$$

  - Groundwater tends to have high hardness

Reducing the effectiveness of soap



Scaling ($CaCO_3$, $Mg(OH)_2$, …)





Hardness

- Hardness is often expressed in equivalents or equivalent weights

  Equivalent weight (EW) = Atomic or molecular weight / n

  where n is related to the valence or ionic charge

  e.g., **EW_{CaCO3} = 100/2 = 50 g/eq = 50 mg/meq**

  EW_{Ca2+} = 40/2 = 20 mg/meq

  $EW_{CO3(2-)}$ = 60/2 = 30 mg/meq

  EW_{HCO3-} = 61/1 = 61 mg/meq

  EW_{Al3+} = 27/3 = 9 mg/meq

- Hardness is typically expressed as **meq/L** or **mg/L as $CaCO_3$**

EW_{CaCO3} = 50 mg/meq

Hardness

TABLE 3

Equivalent Weights for Selected Ions, and Compounds

Name	Symbol or Formula	Atomic or Molecular Weight	Charge n	Equivalent Weight (mg/meq)
Cations				
Aluminum	Al^{3+}	27.0	3	9.0
Calcium	Ca^{2+}	40.1	2	20.0
Hydrogen	H^{+}	1.0	1	1.0
Magnesium	Mg^{2+}	24.3	2	12.2
Potassium	K^{+}	39.1	1	39.1
Sodium	Na^{+}	23.0	1	23.0
Anions				
Bicarbonate	HCO_3^-	61.0	1	61.0
Carbonate	CO_3^{2-}	60.0	2	30.0
Chloride	Cl^-	35.5	1	35.5
Hydroxide	OH^-	17.0	1	17.0
Nitrite	NO_2^-	46.0	1	46.0
Nitrate	NO_3^-	62.0	1	62.0
Orthophosphate	PO_4^{3-}	95.0	3	31.7
Sulfate	SO_4^{2-}	96.0	2	48.0
Neutrals				
Calcium carbonate	$CaCO_3$	100.0	2	50.0
Carbon dioxide	$CO_{2(aq)}$	44.0	2	22.0
Magnesium hydroxide	$Mg(OH)_2$	58.3	2	29.2

Table 6.3, textbook p304

Example 6.4 Expressing the concentration of hardness

A sample of groundwater has 100 mg/L of Ca^{2+} and 10 mg/L of Mg^{2+}. Express its hardness in units of meq/L and mg/L as $CaCO_3$.

Solution:

As meg/L of X = [X]/EW of X, so the contribution of Ca^{2+} in meq/L is:

$$\frac{100\ \text{mg/L}}{20.0\ \text{mg/meq}} = 5.0\ \text{meq/L}$$

The Ca concentration in mg/L as $CaCO_3$ is:

$$\frac{100\ \text{mg/L} \times 50.0\ \text{mg}\ CaCO_3\text{/meq}}{20.0\ \text{mg/meq}} = 250\ \text{mg/L as}\ CaCO_3$$

Similarly, the Mg^{2+} hardness is calculated as 0.82 meq/L and 41.0 mg/L as $CaCO_3$. So, the total hardness is:

5.0 + 0.82 = 5.82 meq/L or 250+41=291 mg/L as $CaCO_3$.

Hardness

Hardness classification of water

Description	meq/L	mg/L as $CaCO_3$
Soft	< 1	< 50
Moderately hard	1-3	50-150
Hard	3-6	150-300
Very hard	> 6	> 300

Table 6.4, textbook p305

USGS general guidelines (http://water.usgs.gov/owq/hardness-alkalinity.html):
Soft: 0 to 60 mg/L as calcium carbonate
Moderately hard: 61 to 120 mg/L
Hard: 121 to 180 mg/L
Very hard: > 180 mg/L as very hard

Alkalinity

- Definition
  - The ability of a water to consume H^+ without significant pH change
  - or, simply H^+ neutralization capability
- Alkalinity of natural water is dominated by carbonate system:

$$\underline{\mathbf{ALK = [HCO_3^-] + 2[CO_3^{2-}] + [OH^-] - [H^+]}}$$

equivalence

$$\underline{\mathbf{ALK = (HCO_3^-) + (CO_3^{2-}) + (OH^-) - (H^+)}}$$

Example 6.5 Calculating alkalinity

A sample of water at pH 10 has 32.0 mg/L of CO_3^{2-}. Find the alkalinity as $CaCO_3$.

Solution:

The EW of CO_3^{2-} is 30 mg/meq. Converting 32.0 mg/L of CO_3^{2-} into mg/L as $CaCO_3$ gives:

$$(CO_3^{2-}) = \frac{32.0\ \text{mg/L} \times 50.0\ \text{mg}\ CaCO_3/\text{meq}}{30.0\ \text{mg/meq}} = 53.3\ \text{mg/L as}\ CaCO_3$$

Equilibrium between carbonate and bicarbonate is reached very rapidly, so the concentration of bicarbonate can be calculated knowing that concentrations in equilibrium expressions must be expressed in mol/L:

$$[HCO_3^-] = \frac{[CO_3^{2-}] \times 10^{-\text{pH}}}{K_2} = \frac{(32/60) \times 10^{-10}}{4.68 \times 10^{-11}} = 1.140\ \text{mmol/L}$$

Because the ionic charge of bicarbonate is 1, 1.140 mmol/L HCO_3^- equals 1.14 meg/L HCO_3^-. Therefore, the concentration of bicarbonate in mg/L as $CaCO_3$ is:

$$(HCO_3^-) = 1.14\ \text{meq/L} \times 50.0\ \text{mg}\ CaCO_3/\text{meq} = 57.0\ \text{mg/L as}\ CaCO_3$$

The pH is 10, so $[H^+]=1\times10^{-10}$ mol/L, and its EW is 1 mg/,eq, so it is:

$$(H^+) = 1\times10^{-10}\text{mol/L} \times 1\ \text{g/mol} \times \frac{10^3\ \text{mg/g}}{1\ \text{mg/meq}} \times 50.0\ \text{mg}\ CaCO_3/\text{meq} = 5\times10^{-6}\ \text{mg/L as}\ CaCO_3$$

Because $[H^+][OH^-]=1\times10^{-14}$, then $[OH^-]= 1\times10^{-4}$ mol/L. The hydroxyl concentration is therefore:

$$(OH^-) = 1\times10^{-4}\text{mol/L} \times 17\ \text{g/mol} \times \frac{10^3\ \text{mg/g}}{17\ \text{mg/meq}} \times 50.0\ \text{mg}\ CaCO_3/\text{meq} = 5\ \text{mg/L as}\ CaCO_3$$

The total alkalinity is:

$$\text{Alkalinity} = (CO_3^{2-}) + (HCO_3^{-}) + (OH^{-}) - (H^{+}) = 53.3 + 57.0 + 5.0 - 5.0\times10^{-6} = 115.3\text{mg/L as } CaCO_3$$

Questions in past exam

A groundwater in a confined aquifer has a pH of 7.8. Its equilibrium partial pressure of CO_2 is 0.01 atm. Estimate the alkalinity of the ground water (in unit of meq/L)

*Henry's law constant K_h=0.0334 mol/(L*atm) at 25 °C, pk_1 = 6.3, pk_2= 10.3

Hardness & Alkalinity

- Total hardness (TH) = $Ca^{2+} + Mg^{2+}$

= **carbonate** hardness (CH) + non-carbonate hardness (NCH)

Associated with **HCO_3^-** and **CO_3^{2-}**

Associated with other anions

If concentration of HCO_3^- and CO_3^{2-} equals or exceeds TH (in essence, alkalinity exceeds TH), then CH = TH & NCH = 0.
If alkalinity less than TH, then CH = alkalinity & NCH = TH-CH.

CH can lead to scaling:

$$Ca^{2+} + 2HCO_3^- \text{ <--> } CaCO_3(s) + CO_2 + H_2O$$

Read Ex. 6.6

Softening

- Reducing water hardness
  - Surface water, hardness typically < 200 mg/L as CaCO3 → softening not required
  - Groundwater, common to have high hardness
- Methods
  - Lime-soda
    - adding lime to adjust pH > 10
    - adding lime $Ca(OH)_2$ for CH removal
    - adding soda Na_2CO_3 for NCH removal
    - Recarbonation (adding $CO_{2(g)}$) is needed to adjust pH
  - Ion exchange
    - Provide higher quality water than lime-soda method
    - More expensive

Lime-soda Softening

1. Removal of $CO_{2(aq)}$:

$$CO_{2(aq)} + Ca(OH)_2 \leftrightarrow CaCO_{3(s)} + H_2O \quad (6.37)$$

2. Removal of calcium carbonate hardness (Ca-CH):

$$Ca^{2+} + 2HCO_3^- + Ca(OH)_2 \leftrightarrow 2CaCO_{3(s)} + 2H_2O \quad (6.38)$$

3. Removal of magnesium carbonate hardness (Mg-CH):

$$Mg^{2+} + 2HCO_3^- + 2Ca(OH)_2 \leftrightarrow 2CaCO_{3(s)} + Mg(OH)_{2(s)} + 2H_2O \quad (6.39)$$

4. Removal of calcium noncarbonate hardness (Ca-NCH):

$$Ca^{2+} + Na_2CO_3 \leftrightarrow CaCO_{3(s)} + 2Na^+ \quad (6.40)$$

5. Removal of magnesium noncarbonate hardness (Mg-NCH):

$$Mg^{2+} + Ca(OH)_2 + Na_2CO_3 \leftrightarrow CaCO_{3(s)} + Mg(OH)_{2(s)} + 2Na^+ \quad (6.41)$$

Textbook, p310

Groundwater Treatment Systems

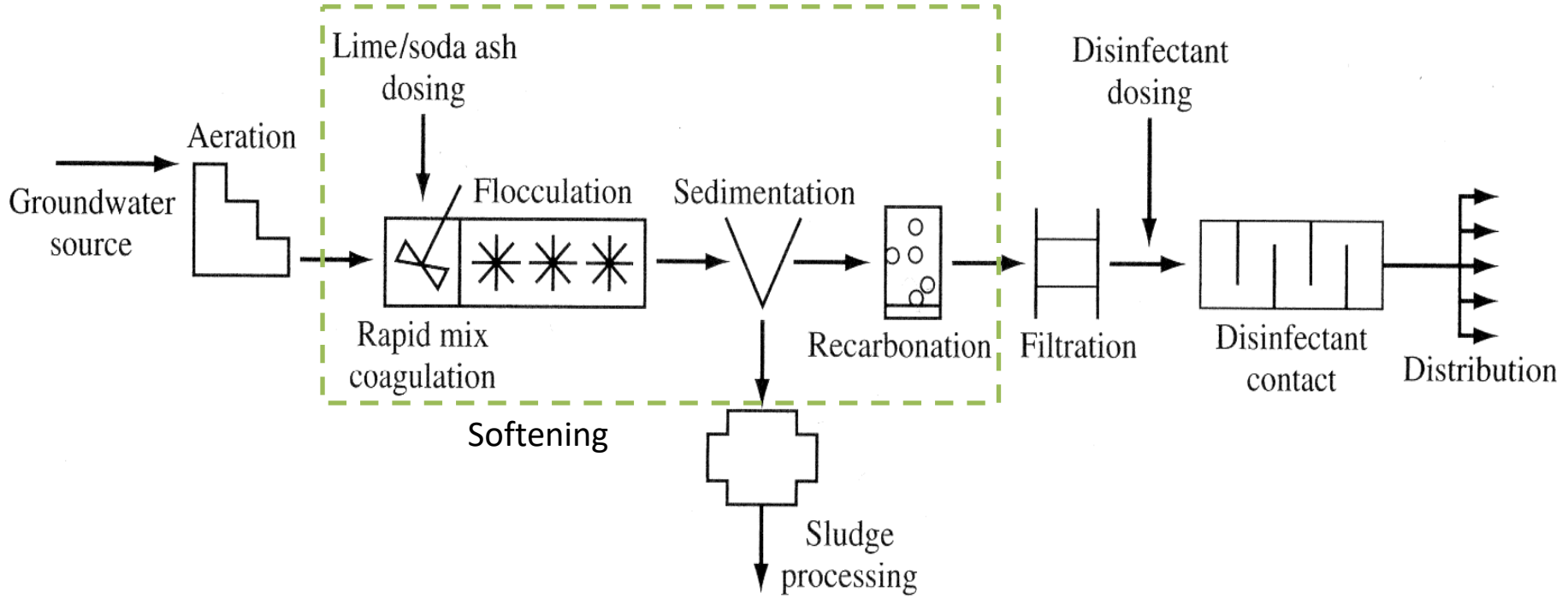
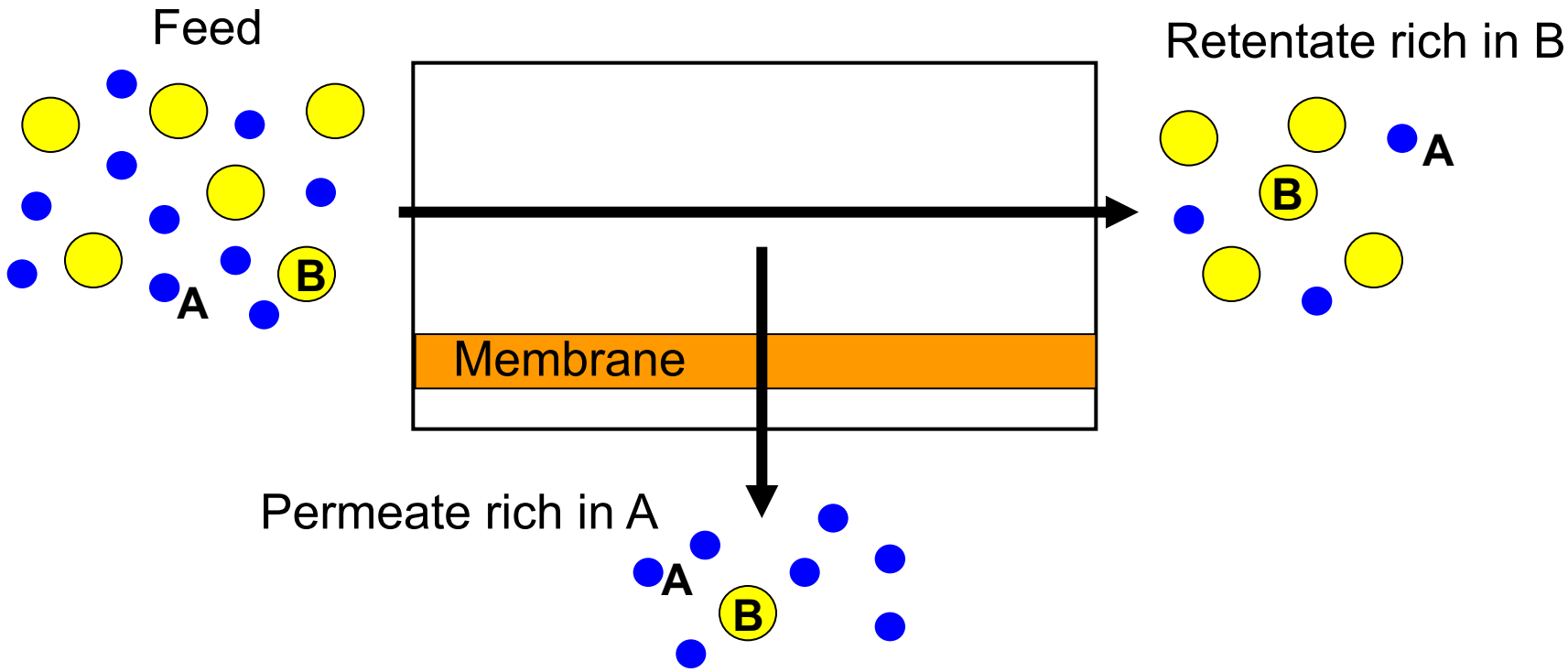


FIGURE 6.3 Schematic of a typical water treatment plant for groundwater, including softening for calcium and magnesium removal.

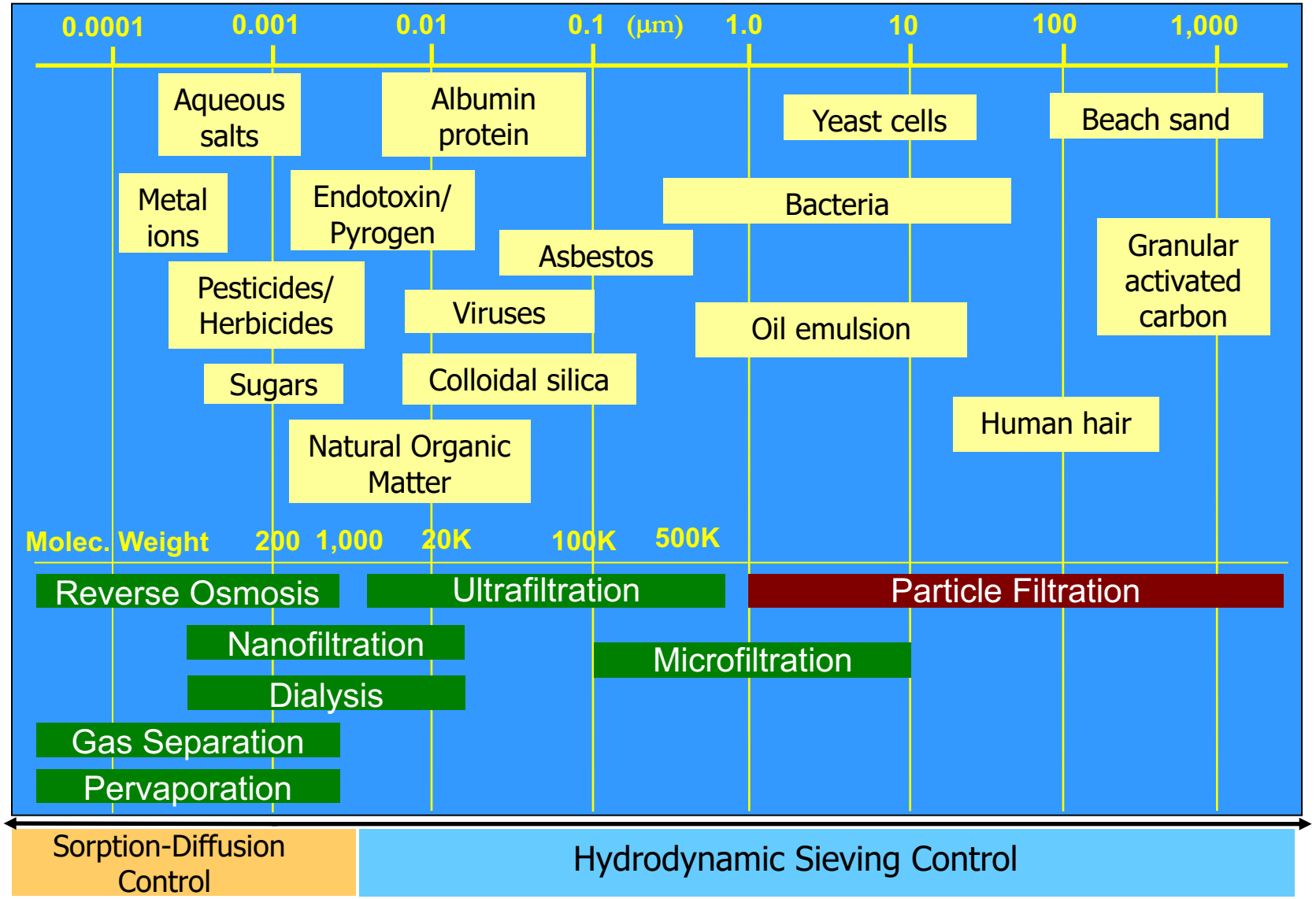
Textbook, Figure 6.3

Membrane processes

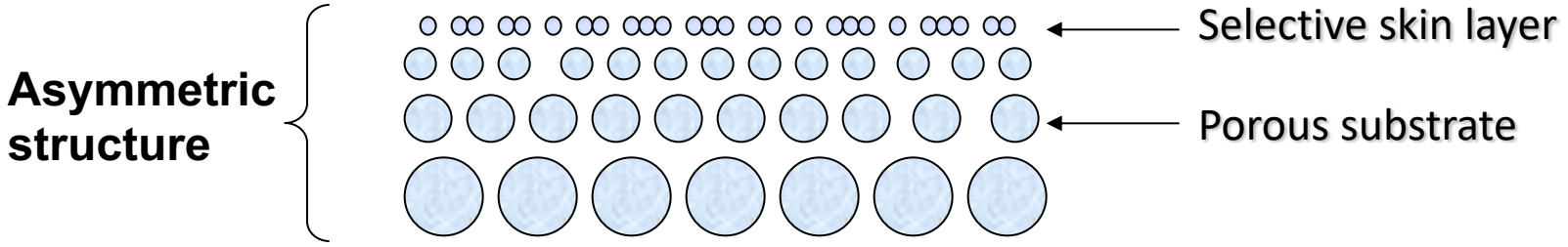
A **selective barrier** that allows the **separation** of components in a mixture.



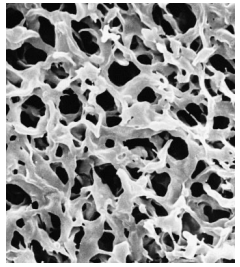
Membrane processes

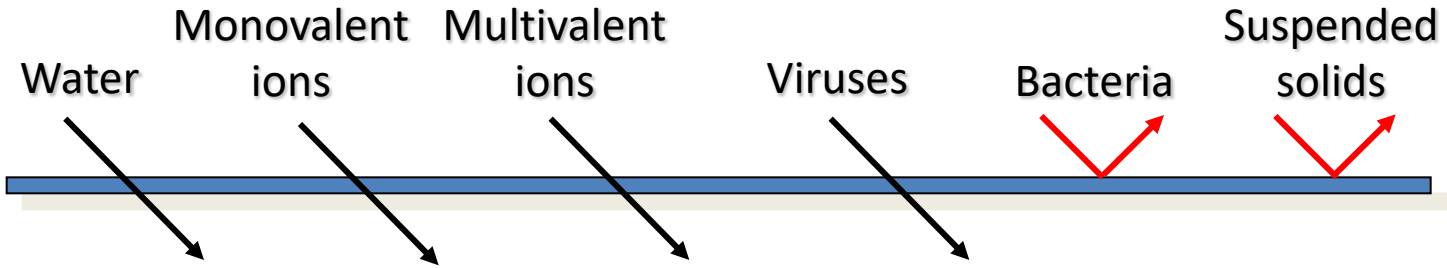


Membrane processes

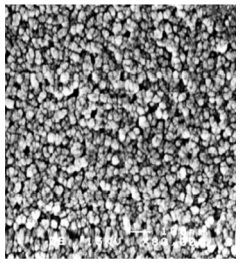


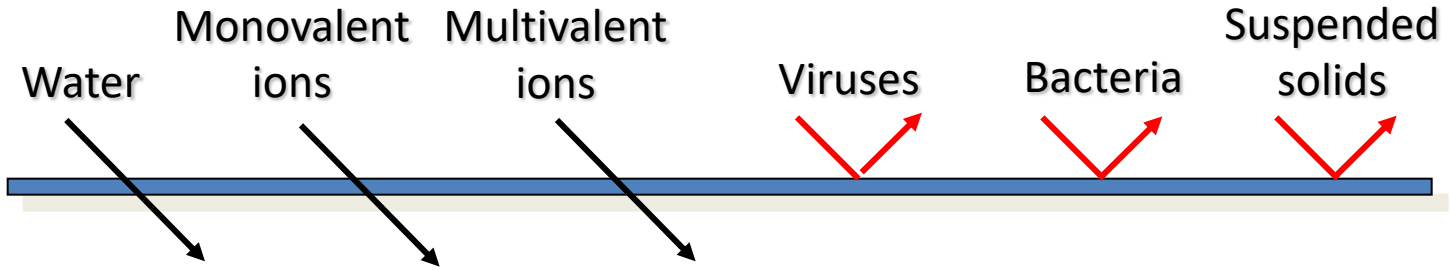
Microfiltration



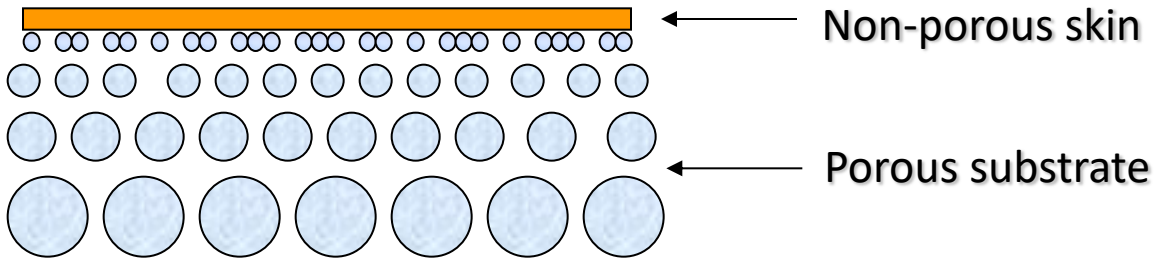


Ultrafiltration

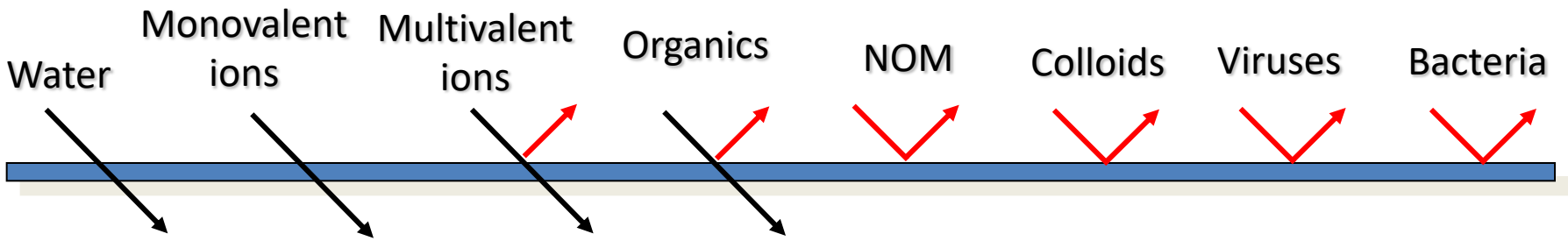




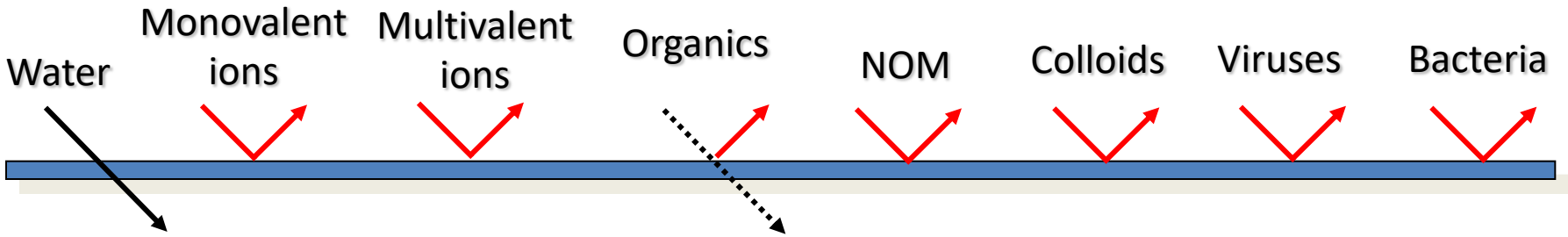
Membrane processes



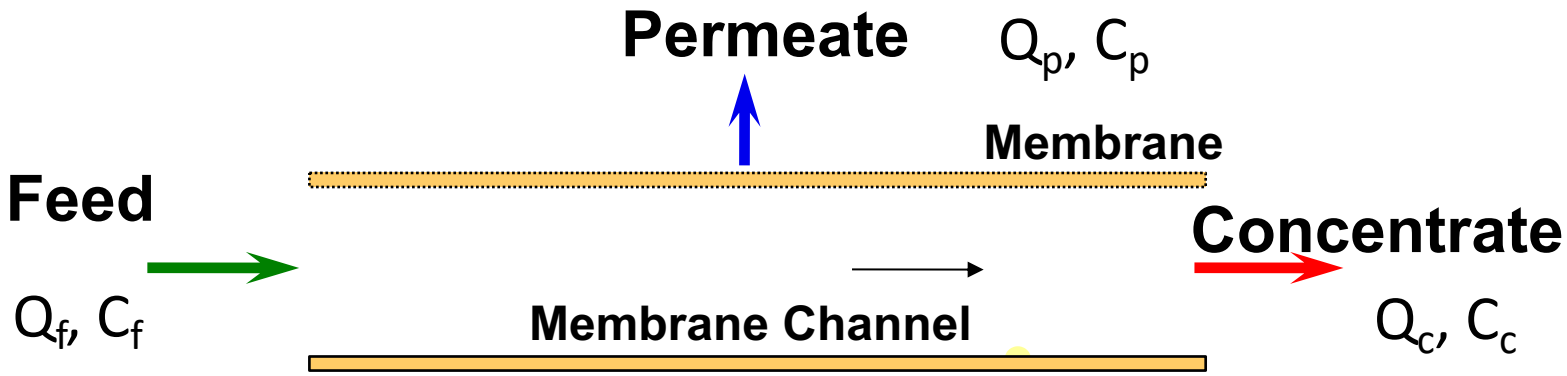
Nanofiltration



Reverse Osmosis



Membrane processes



- Feed (feedwater): the raw water that enters the membrane separation system
- Permeate: the treated water that passes through the membrane barrier layer
- Retentate (concentrate): the difference between feed and permeate

$$\text{Flux} = Q_p/A_m$$

$$\text{Rejection } R = 1 - C_p/C_f$$

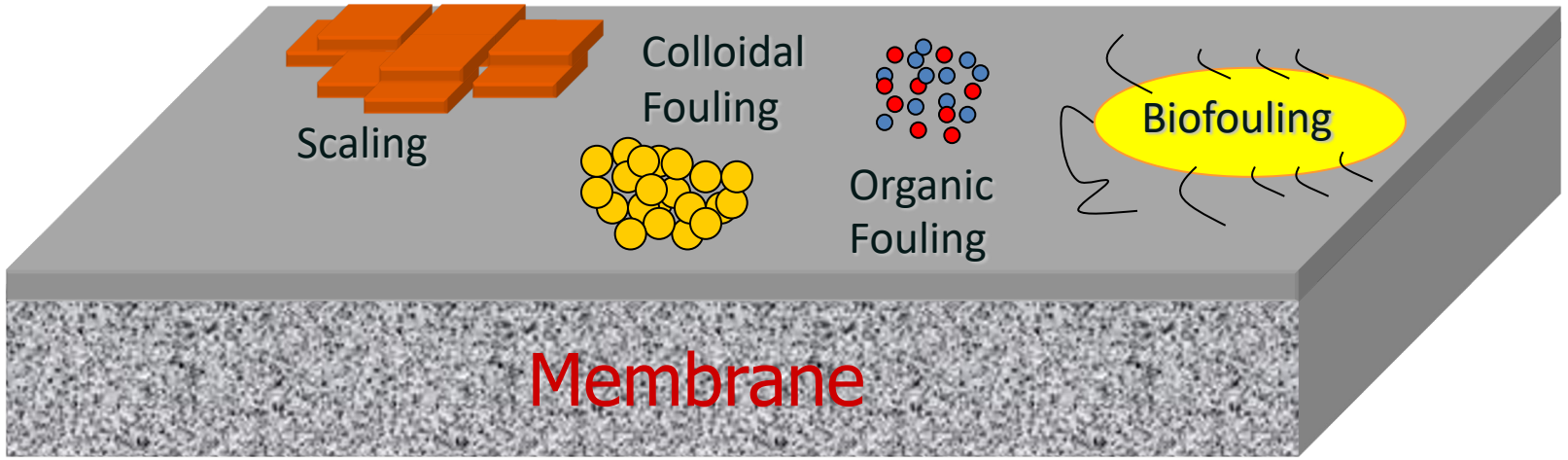
Log removal = -log(1-R)

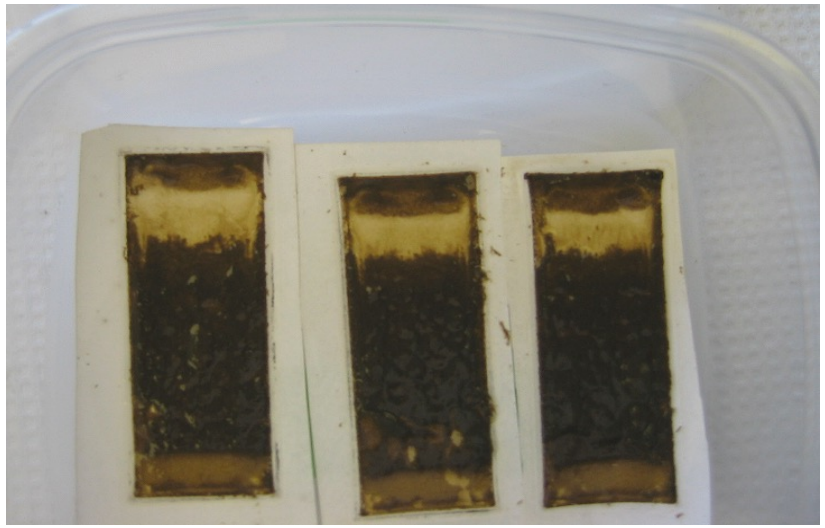
1 log removal → 90% rejection

2 log removal → 99% rejection

$$\text{Recovery} = Q_p / Q_f$$

Membrane fouling

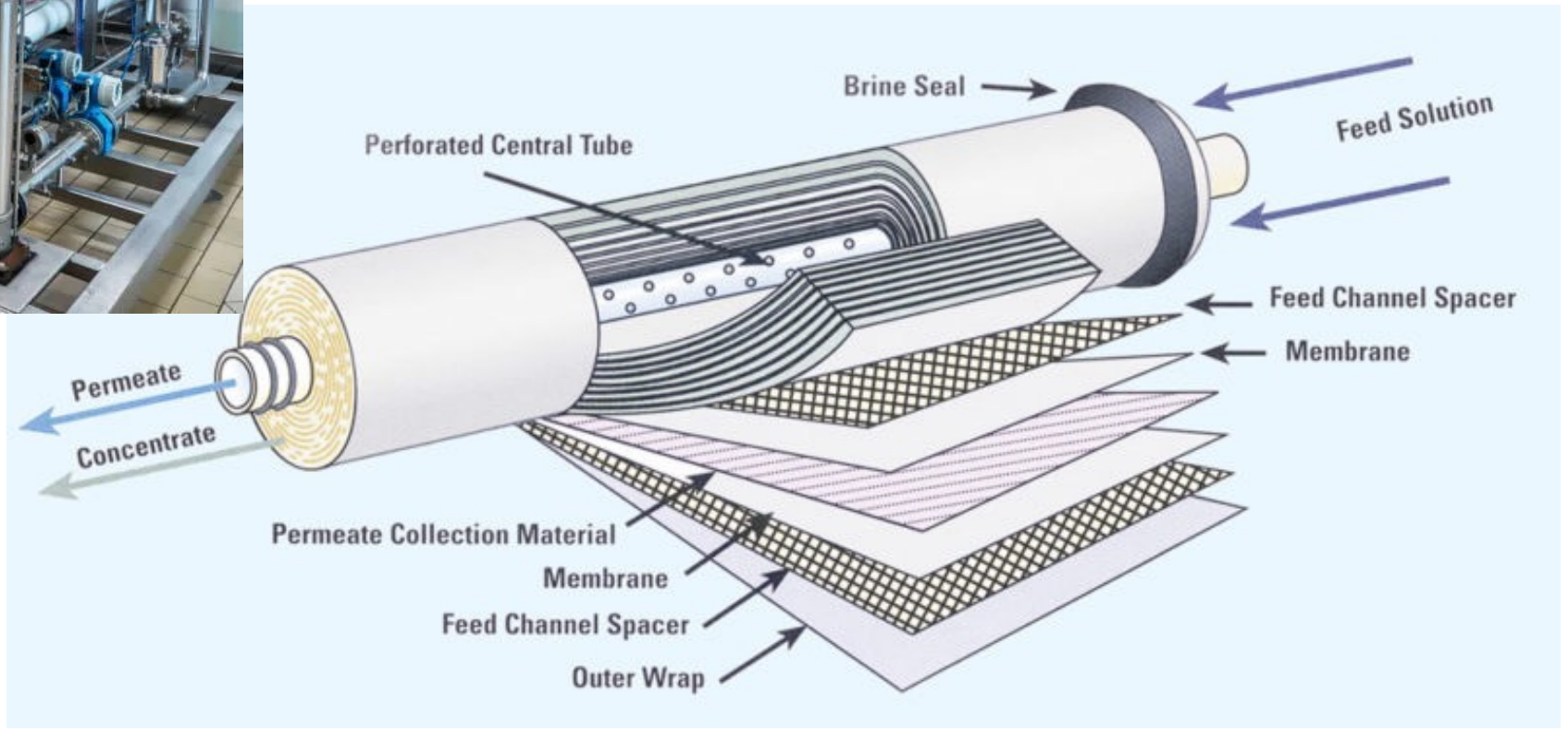




Fouled membranes after 20 days of direct filtration of secondary wastewater effluent.

- Increase membrane separation resistances
- Reduce water production
- Reduce/alter membrane selectivity
- Shorten membrane lifespan
- Increase operational cost




Brine Seal
Feed Solution
Perforated Central Tube
Feed Channel Spacer
Membrane
Permeate
Concentrate
Permeate Collection Material
Membrane
Feed Channel Spacer
Outer Wrap

Questions in past exam

An inland city has an available water source containing 5 mmol/L of Ca^{2+}, 0.2 mmol/L of SO_4^{2-}, 1 mmol/L of Mg^{2+}, 12 mmol/L of Cl^-, and other dissolved ions.

(a) Estimate the hardness of the water in mg/L as $CaCO_3$.

(b) If reverse osmosis membrane is used to treat the water in the question, briefly discuss the potential problems.

Readings and practice questions

- Readings
  - Essential reading: pp281-317

- Practice questions
  - Q6.4, 6.8, 6.11, 6.20, 6.24, 6.25